

Research Paper

Extensive analysis of hydrodynamics in liquid–solid fluidized beds with particle-resolved simulations

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ABSTRACT

Particle-Resolved Simulations (PRS) of fluid-solid particles are performed to investigate the hydrodynamics of liquid–solid fluidized bed. The various physical phenomenon contributing to turbulence generation in fluidized beds are first analyzed. A liquid–solid fluidized bed reproducing the experimental configuration of Corona et al. (2011) is then investigated. Four fluidization velocities are considered, corresponding to solid volume fractions ranging from 0.13 to 0.35. Grid sensitivity is examined with 12 and 24 grid cells per particle diameter, and a refined resolution of 36 cells per diameter is investigated at the lowest fluidization velocity. Particle and fluid statistics are computed using both Lagrangian and Eulerian approaches. The results show that the bed expansion height is accurately predicted even with the coarsest grid resolution. However, higher spatial resolutions are required to correctly capture the fluctuating kinetic energy profiles and the autocorrelation of the vertical fluid velocity. Finally, an extension of the correlation proposed by Alm  ras et al. (2019) is introduced to predict the fluid velocity variance as a function of the mean fluid velocity, solid volume fraction, and Stokes number based on the terminal settling velocity. The proposed correlation is validated for Stokes numbers up to 7.

1. Introduction

Many of the physical mechanisms involved in fluidized beds are also present in other types of fluid-particle flows as swarms of bubbles or droplets. Two-phase flows composed of a continuous phase and a dispersed phase are widespread in industry and in nature, which has attracted the interest of the scientific community. Describing the dynamic of these flows is challenging because of the superposition of a multitude of physical contributions. As discussed by Risso (2018), the disturbance of the mean flow can be explained on the one hand by the turbulence induced by the shear stress of the fluid or *Shear Induced Turbulence* (SIT), and on the other hand by the agitation induced by the presence of particles, also called pseudo-turbulence. As the author explains, when observed from the reference frame of a particle (or bubble), the disturbance of the flow caused by the presence of the particles generates spatial fluctuations, but not temporal fluctuations as turbulence would. The term is commonly referred to as *Bubble Induced Agitation* (BIA) or *Bubble Induced Fluctuations* (BIF) to avoid any misleading with turbulence. Numerous studies have extensively characterized BIF and SIT for liquid-gas flows (Riboux et al., 2013; Risso, 2016; Lance and Bataille, 1991; Cluzeau et al., 2022) and for fluid-solid particles flows (Amoura et al., 2017; Alm  ras et al., 2019,

2021). However, the link between the fluid and the particles agitation has not yet been fully characterized (Alm  ras et al., 2021).

Therefore, before studying the dynamics of fluidized beds, it is important to understand the various interactions that occur within fluid-particle flows. Two main mechanisms are involved: SIT and BIA (or solid particle-induced agitation) (Risso, 2016; Amoura et al., 2017) - also referred as BIF (Cluzeau, 2019) or pseudo-turbulence (Lance and Bataille, 1991; Alm  ras et al., 2019). While the various phenomena concern bubble flows, the same mechanisms apply to fluid-solid flows (Risso et al., 2008; Amoura et al., 2017). Slight differences between the two types of flows, resulting from different boundary conditions at the interfaces, are observed in the shape of the PDFs of velocity fluctuations (Amoura et al., 2017). In the case of bubble flows, the movement of the fluid and bubbles can be caused by an external pressure gradient or by buoyancy. While the description of interactions presented above is appropriate for bubble flows, the same type of interactions occur in fluid-solid flows. Characterizing each contribution is particularly challenging due to their interdependence. Moreover, BIF itself arises from two distinct contributions (Risso et al., 2008; Risso, 2019). The first is a result of the average disturbance generated near the particles by the presence of the wake and potential flow in the vicinity of the particles. The second contribution is the shear-induced

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Nomenclature

Roman symbols

C_D^f	Drag coefficient of an isolated particle [–]
D_{ii}	Self-diffusion coefficient in direction i [$\text{m}^2 \text{s}^{-1}$]
D_h, D_v	Self-diffusion coefficient in the horizontal/vertical direction [$\text{m}^2 \text{s}^{-1}$]
D_p	Particle diameter [m]
Δt_{avg}	Simulation duration over which statistics are averaged [s]
Δt_{simu}	Simulation time step [s]
Δx	Eulerian grid cell size [m]
E_p, E_f	Fluctuating kinetic energy of particles/fluid [$\text{m}^2 \text{s}^{-2}$]
F_c	Collision source term in the momentum equation [N m^{-3}]
$G(r)$	Radial pair distribution function [–]
$\mathbf{g}$	Gravity vector [m s^{-2}]
h_{max}	Bed height [m]
I	Fluid volume fraction [–]
k_{anis}	Velocity variance anisotropy coefficient [–]
$M_{ii}(t)$	Variance of particle displacement in direction i [m^2]
N_p	Number of particles [–]
N_t	Number of time steps considered [–]
p	Pressure [Pa]
r_{ij}	Gravity centers distance of particles i and j [m]
Re_t	Reynolds number based on the terminal settling velocity [–]
$R_{ii}(\tau)$	Autocorrelation coefficient of the i th velocity component [–]
St_t	Stokes number based on the terminal velocity [–]
t	Time [s]
$t_{L,ii}$	Lagrangian diffusion timescale in the direction i [s]
$t_{L,v}, t_{L,h}$	Lagrangian timescale in the vertical/horizontal direction [s]
$\mathbf{u}$	Velocity vector [m s^{-1}]
U_F	Fluidization velocity [m s^{-1}]
U_t	Terminal settling velocity of an isolated particle [m s^{-1}]

Greek symbols

α	Particle volume fraction [–]
γ_i	Proportionality constant in the Alm��ras fluid velocity variance model, direction $i \in \{x, z\}$ [–]
ζ_i	Proportionality constant in the extended correlation [–]
μ_f	Fluid dynamic viscosity [Pa s]
μ_p	Particle numerical viscosity [Pa s]
ρ_f	Fluid density [kg m^{-3}]
ρ_p	Particle density [kg m^{-3}]
τ	Time lag for velocity autocorrelation [s]
τ_f	Characteristic time of fluid motion at the particle scale [s]
τ_{fp}	Particle relaxation time [s]
χ	Phase indicator function [–]

Subscripts and superscripts

A	Configuration of Alm��ras et al. (2019)
C	Configuration of Corona et al. (2011)
E	Eulerian quantity
f	Fluid phase
i, j, k	Generic direction index (x, y, z) or particle index
L	Lagrangian quantity
p	Particle
t	Terminal settling (as in U_t, Re_t, St_t)
v, h	Vertical/horizontal (as in $t_{L,v}, t_{L,h}, D_v, D_h$)
x, z	Horizontal directions (transverse)
y	Vertical direction (gravity-aligned)

Operators and notation

$\langle \cdot \rangle_{i,j}$	Spatial average in directions i and j
$\langle \cdot \rangle_p$	Ensemble average over all particles
∇	Nabla operator for divergence or gradient
$\otimes$	Tensor (outer) product
$\bar{\cdot}$	Temporal average of quantity u
$\sim$	Ensemble average

Abbreviations and acronyms

BIA	Bubble-Induced Agitation (synonym of BIF)
BIF	Bubble-Induced Fluctuations
BIT	Bubble-Induced Turbulence
DEM	Discrete Element Method
FAC	Self-diffusion coefficient computed from the autocorrelation function
PDF	Probability Density Function
PRS	Particle-Resolved Simulation
SIT	Shear-Induced Turbulence
SPT	Single-Phase Turbulence
VD	Self-diffusion coefficient computed from the variance of displacement
WIF	Wake-Induced Fluctuations (spatial component of BIF)
WIT	Wake-Induced Turbulence (temporal component of BIF)

turbulence that develops when the Reynolds number is sufficiently high. The turbulence then results from collective instability and the interaction of particle wakes. Working in the reference frame of bubbles rising in the fluid at the same velocity, Riboux et al. (2013) decomposed the fluid velocity fluctuation into a spatial contribution and a temporal contribution:

$$\hat{u}(x, t) = \bar{v}(x) + v'(x, t) \quad (1)$$

with $\hat{u}(x, t) = u(x, t) - U_\infty$ where U_∞ represents the bubbles' velocity in the fluid, $\bar{v}(x) = v(x, t)$, $v'(x, t) = 0$. $u(x, t)$ represents the fluid velocity in the reference frame of the experiment. The two velocity components are noted v to avoid confusion with the decomposition in Eq. (2). The operator $\bar{\cdot}$ represents the temporal average. In the case of bubbles with a diameter of 2.5 mm and a presence rate of $\alpha = 0.025$, the authors revealed that the spatial component is predominant for the vertical velocity and is localized in the wake of the particles. The temporal component is very weak compared to the spatial component in this homogeneous case. For horizontal velocity, fluctuations are weaker and the contribution of the average wake is negligible, as are spatial fluctuations throughout the domain. Horizontal velocity fluctuations

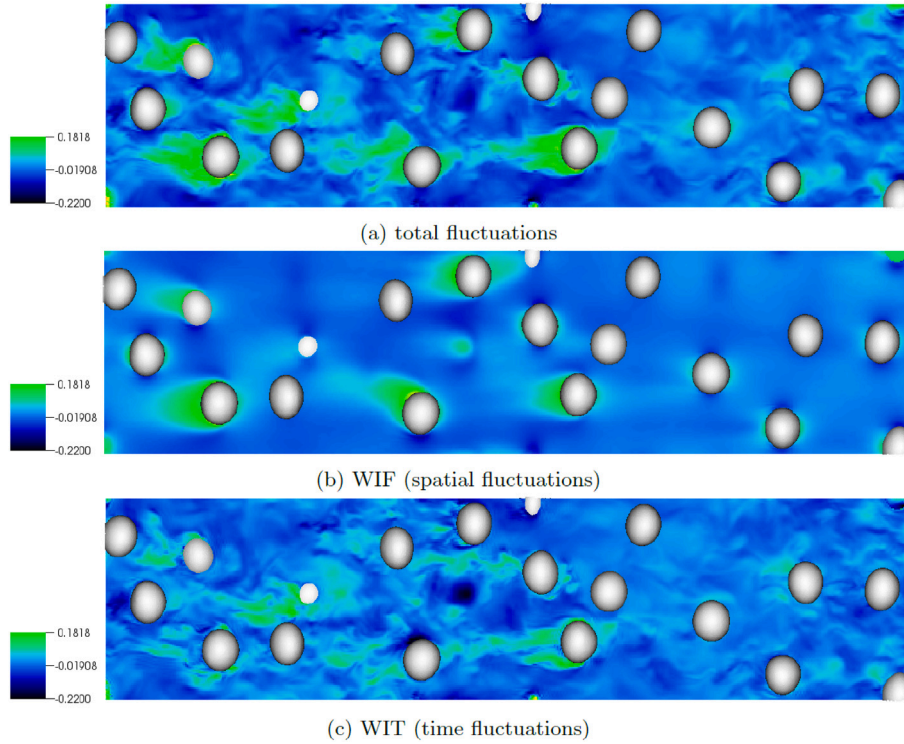


Fig. 1. Visualization of the spatial and temporal components of total turbulent fluctuations.
Source: Image adapted from Cluzeau et al. (2022).

therefore result from temporal fluctuations. A distinction is drawn between the BIF and the BIT (*Bubble-Induced Turbulence*) (Risso, 2019). However, the properties of BIT differ from those of single-phase SIT, because, on the one hand, the variance of vertical velocity fluctuations is greater than that of horizontal velocity, and, on the other hand, the PDFs of velocity fluctuations are anisotropic (Risso et al., 2008). Since BIT can generate SIT, it seems clearer to use other names. Thus, as proposed by Cluzeau (2019), one will not refer to SIT but rather to SPT *Single-Phase Turbulence* and BIF. BIF is the result of an induced spatial fluctuation (referred to as WIF *Wake-Induced Fluctuations*), and a temporal fluctuation (referred to as WIT *Wake-Induced Turbulence*). The decomposition is illustrated in Fig. 1, which was extracted from the study of Cluzeau et al. (2019).

WIF is present for all Reynolds numbers and is responsible for shifting the PDFs of vertical velocity fluctuations toward positive values (Amoura et al., 2017). This phenomenon can be attributed to the fluid movement induced by the wake of the solid particles. It is also responsible for the predominance of vertical velocity fluctuations over horizontal fluctuations. Reynolds numbers greater than 200, based on the particle diameter, are required for WIT to develop (Amoura et al., 2017). Unlike WIF, WIT is characterized by symmetric and almost isotropic PDFs described by exponential laws. This separation of contributions was made possible by the experimental study of a fixed array of spheres randomly distributed throughout the domain. Other configurations allow the two contributions to be separated. Examples include Hele-Shaw cells (Bouche et al., 2012, 2014), within which turbulence (SPT, WIT) cannot develop, or simulations where the particle Reynolds numbers are less than 200 (Cluzeau, 2019). At higher particle concentrations, it becomes difficult to identify the wake of each particle as the wakes overlap (see, for example, Bouche et al. (2014) with Hele-Shaw cells). Furthermore, the wake of particles in a fluidized bed is smaller than that of an isolated particle (see Figure 3 in Bouche et al. (2014)).

In the context of fluidized beds, it is more difficult to separate the different contributions of the BIF. Indeed, the fluid is no longer at

rest and is characterized by an upward average movement in which the particles are moving in chaotic patterns. Therefore, SPT (or WIT) develops at the walls if the Reynolds number is sufficiently large. The agitation can no longer be described using the same decomposition. In fluidized beds, the particle solid fraction reported in the literature are usually higher, between approximately 10% and 40% (Corona, 2008; Ozel et al., 2017; Alm  ras et al., 2021; Gueguen et al., 2021; Hamidi et al., 2023). In this context, the high solid fraction offers an advantage, as turbulence cannot develop in the interstitial area between the fluid and the particles (Alm  ras et al., 2019). WIF thus predominates over SPT and WIT. Risso (2016) describes the fluctuations in the continuous phase by two contributions: the disturbance generated in the vicinity of each particle and the turbulent fluctuations resulting from the instability of the flow through a random array of particles. Based on this description, Alm  ras et al. (2019) proposed a model that relates the variance of the liquid's fluctuating velocity to the variance of the solid fraction. The authors also introduced a model that links the velocity fluctuations of the particles to those of the fluid. It is based on the fact that the instantaneous vertical velocity of a fluid point in the interstitial area is distributed around the average velocity of the liquid. Whereas for a fluid point located in the wake of a particle, the velocity is distributed around the average vertical velocity of the particles. For a fluidized bed (non-transported particles), the average vertical velocity of the particles is zero. Fluctuations of the vertical velocity of the fluid can be defined by:

$$u'_{f,y} = u_{f,y} - \widetilde{U_{f,y}} \quad (2)$$

where $\widetilde{\cdot}$ is the ensemble average operator, equivalent to a spatial average or a temporal average under the assumption of spatial and temporal ergodicity. These fluctuations are typically on the order of $-\widetilde{U_{f,y}}$ in the wake of the particles (since $u_{f,y} \sim \widetilde{u_{p,y}} = 0$ in this case) and depend on the local solid fraction. After a few lines of development and using the Richardson-Zaki correlation (Richardson and Zaki, 1954), the authors obtain the following equations:

$$\widetilde{u'^2_{f,i}} = \gamma_i^2 \widetilde{U_{f,y}^2} \alpha (1 - \alpha) \quad (3)$$

$$\overline{u_{p,y}^2} = \overline{u_{f,y}^2} \left(1 + \frac{k_y}{\overline{u_{f,y}^2}^{1/2}} \frac{d(|\overline{U_{f,y}} - \overline{U_{p,y}}|)}{d\alpha} [\alpha(1-\alpha)]^{1/2} \right)^2 \quad (4)$$

$$\overline{u_{p,x}^2} = \overline{u_{f,x}^2} \left(1 + \frac{k_x}{\overline{u_{f,x}^2}^{1/2}} \frac{d(|\overline{U_{f,x}} - \overline{U_{p,x}}|)}{d\alpha} [\alpha(1-\alpha)]^{1/2} \right)^2 \quad (5)$$

where $i \in \{x, z\}$, $\overline{U_{f,y}} = \frac{U_F}{1-\alpha}$ and $\overline{U_{p,y}} = 0$. γ_i , k_x and k_y are constants to be determined. In the above equations, y represents the vertical component and x - or z - the horizontal one. Their model is validated by confrontation to an experimental study for solid fraction between 10% and 45%. This model was then extended to the analysis of PDFs of velocity fluctuations (Alm  ras et al., 2021).

Ozel et al. (2017) and Hamidi et al. (2023) numerically reproduced Corona's liquid fluidized bed experiment (Corona, 2008). The volume fraction of particles ranged from 10% to 40%. The authors studied the statistics of particles' velocity (Ozel et al., 2017; Hamidi et al., 2023) and the turbulent kinetic energy of the fluid (Ozel et al., 2017). In line with the experimental results, the authors showed that the variance of particles' velocity decreases with the solid fraction for the vertical and horizontal components. Moreover, the anisotropy of the velocity variance is constant over the range of presence rates studied. The turbulent kinetic energy of the fluid is greater than that of the particles (by a ratio of at least two). Furthermore, the ratio E_f/E_p increases with the presence rate. Finally, the time scales of particle velocity fluctuations were studied by analyzing the autocorrelation functions of particle velocities. In the present work, the experimental study of Corona (2008) is reproduced numerically. An extensive analysis of the particle and fluid statistics is performed, followed by an extension of the correlation developed by Alm  ras et al. (2019). The new correlation is then validated against a different set of simulations of fluidized beds.

The paper is organized as follows. First, the numerical and physical modeling are described in Section 2. The fluidized bed of Corona (2008) and some key aspects of the numerical analysis are presented in Section 3. The analysis of the statistics of the fluid and the particles are described in Sections 4 and 5. Section 6 introduces the extension of the model of Alm  ras et al. (2019). Finally, conclusions and outlooks are drawn in Section 7.

2. Physical and numerical modeling

The fluidized bed simulations presented in this paper were performed using TrioCFD software. TrioCFD is developed by the French atomic energy commission (CEA) since 1993 (Calvin et al., 2002). This software was previously used to study turbulence (Benarafa et al., 2006), anisothermal bubbly flows (Grosso et al., 2024), fluid-solid flows (Hamidi et al., 2023; Butaye et al., 2023) and anisothermal fluid-solid flows (Butaye et al., 2025). Version 1.9.1 of TRUST/TrioCFD was used to investigate liquid fluidized beds, which was the most recent available version at the beginning of the study. The numerical modeling of the particle-resolved approach is described in Butaye et al. (2025). The key elements of the method are briefly exposed below.

The model is based on a one-fluid formulation of the Navier–Stokes equations. Each phase are assumed incompressible and particles are spherical and non-deformable. Fluid and particles physical properties are constants. Navier–Stokes equations are solved on an Eulerian grid to predict the fluid flow. Particles are described by a mobile Lagrangian mesh, superimposed on the Eulerian mesh. Lagrangian markers that describe each particle are advected by the resolution of a transport equation. This method is referred to as the Front-Tracking method, originally developed by Unverdi and Tryggvason (1992). To enforce the solid-body motion, all lagrangian markers of a particle are transported with the same velocity. Furthermore, the viscous penalty method (Ritz and Caltagirone, 1999; Vincent et al., 2014) is employed to implicitly enforce fluid constraints at the fluid/solid interface. The soft-sphere

collision developed in Hamidi et al. (2023) models the particle–particle and wall-particle collision process.

The fluid flow is governed by the following mass and momentum equations:

$$\nabla \cdot \mathbf{u} = 0 \quad (6)$$

$$\frac{\partial(\rho \mathbf{u})}{\partial t} + \nabla \cdot (\rho \mathbf{u} \otimes \mathbf{u}) = -\nabla p + \nabla \cdot (\mu (\nabla \mathbf{u} + \nabla^T \mathbf{u})) + \rho \mathbf{g} + \mathbf{F}_c \quad (7)$$

$\mathbf{u}$ is the velocity vector, p the pressure, t the time, $\mathbf{g}$ the gravity vector, ρ the density, μ the viscosity. The definition of ρ and μ in the two-phase eulerian cells is given by Eqs. (11) and (12) respectively. A collision source term $\mathbf{F}_c$ is added to the classical formulation of the momentum equation to consider particle–particle and wall-particle collisions. $\mathbf{F}_c$ links the collision force computed on a Lagrangian manner and the volumic force applied on the Eulerian field on which Navier–Stokes equations are solved. Velocity and pressure are solved on a staggered grid. Scalar quantities are computed at the center of Eulerian cells whereas vector quantities are computed at the center of the faces. The solid behavior of particles imposes a zero deformation tensor inside their volume. This condition is fulfilled by imposing a high viscosity ratio between the fluid and the particles (viscous penalty method Ritz and Caltagirone, 1999; Vincent et al., 2014). The interface is transported by a transport equation:

$$\frac{\partial \chi}{\partial t} + \mathbf{u} \cdot \nabla \chi = 0 \quad (8)$$

where χ is the phase indicator function. $\chi = 0$ in the solid phase and $\chi = 1$ in the fluid phase. Numerically, the Eulerian velocity is interpolated at each Lagrangian marker. Then, the velocity is computed at the gravity center of each Lagrangian facets as the average of the velocity of the three lagrangian markers. The velocity of the particle is computed as a surfacic average of the velocity of each lagrangian facet. Finally, each Lagrangian marker is transported with the velocity $\mathbf{u}_p$:

$$\mathbf{x}_i^{t+\Delta t} = \mathbf{x}_i^t + \mathbf{u}_p \Delta t \quad (9)$$

where $\mathbf{x}_i$ is the position of the Lagrangian marker i and Δt is the time step of the simulation. The Lagrangian markers are not physical points of the interface (Butaye et al., 2023). The projection of the particle's surface onto the Eulerian grid remains unchanged with pure rotation. Therefore, the rotation of the Lagrangian markers relative to the particle's gravity center is not considered.

In two-phase cells, the physical properties of the fluid are defined by the volumic solid fraction. In each Eulerian cell of volume V , the volume phase indicator function for the fluid is defined as:

$$I = \int_V \chi dV \quad (10)$$

The density is computed with an arithmetic average (see Eq. (11)) whereas the viscosity is computed with a harmonic average (Toutant, 2025; Magnaudet et al., 2025) (see Eq. (12)).

$$\rho = I\rho_f + (1-I)\rho_p \quad (11)$$

$$\mu = \frac{\mu_f \mu_p}{I\mu_p + (1-I)\mu_f} \quad (12)$$

The soft-sphere collision model of Hamidi et al. (2023) is used for particle–particle and wall-particle interactions. The reader is referred to Hamidi et al. (2023) and Butaye et al. (2025) for more details on the computation method.

3. Description of the studied liquid–solid fluidized bed

The case study, an isothermal liquid–solid fluidized bed of 2134 particles, is described in Section 3.1. The macroscopic behavior of the fluidized bed is presented in Section 3.2. Bed height, solid fraction are analyzed to assess the fluidization law of Richardson and Zaki (1954). Then, the effect of the grid resolution on the solid fraction is detailed in Section 3.3.

Table 1
Numerical parameters.

Velocity	Dimensions (L/D_p)	Grid resolution ($D_p/\Delta x$)	Number of CPUs
$U_F < 0.15 \text{ m s}^{-1}$	$12 \times 54 \times 12$	12	288
	$12 \times 27 \times 12$	24	1152
	$12 \times 27 \times 12$	36	3888
$U_F = 0.15 \text{ m s}^{-1}$	$12 \times 108 \times 12$	12	544

3.1. Case study: isothermal liquid–solid fluidized bed

The fluidized bed simulations performed in this work reproduce the experimental study by Corona (2008) and Corona et al. (2011). The authors studied a liquid fluidized bed in a vertical cylindrical tube with a diameter of 8 cm and a height of 60 cm. The fluid is an aqueous solution of potassium thiocyanate with a viscosity of $\mu_f = 3.8 \cdot 10^{-3} \text{ Pa s}$ and a density of $\rho_f = 1400 \text{ kg m}^{-3}$. The particles are Pyrex spheres with a diameter of $D_p = 6 \text{ mm}$ and a density of $\rho_p = 2230 \text{ kg m}^{-3}$. Numerically, a viscosity ratio of 10^4 was applied for the viscous penalty method. The particles therefore have a numerical viscosity of $\mu_p = 38 \text{ Pa s}$. The number of particles to be injected into the domain was calculated by Hamidi et al. (2023) based on the volume fraction and bed height given by experimental results (Corona, 2008). Thus, 2134 solid particles are considered in these simulations. The simulated domain is a rectangular parallelepiped with an equivalent flow section, as TrioCFD does not enable Finite Volumes studies on cylindrical domains. The square flow section defined in the simulations is 3.1% larger than the experimental cylindrical flow section. A no-slip condition is applied to the side walls of the domain. The fluid is injected from the bottom of the domain at a velocity U_F . An imposed pressure condition is applied to the outlet section. Four fluidization velocities are considered, $U_F \in \{0.073; 0.09; 0.12; 0.15\} \text{ m s}^{-1}$, as well as three grid resolutions, $D_p/\Delta x \in \{12, 24, 36\}$. The grid resolution of 36 meshes per particle diameter was only studied for the lowest fluidization velocity due to the numerical cost of the simulation. The simulation time step, Δt_{simu} , range from $5 \cdot 10^{-5} \text{ s}$ to $1 \cdot 10^{-4} \text{ s}$. To save computational time, the height of the domain was adapted to the inlet velocity and the grid resolution. Numerical parameters of the simulations are given in Table 1.

3.2. Macroscopic description of the fluidized bed

3.2.1. Bed height

The bed height is computed in a Lagrangian manner by determining the position of the highest particle in the domain. The time evolution of the height for the four fluidization velocities is shown in Fig. 2 and compared with the simulation results of Ozel et al. (2017). Since the initialization of particle positions (and thus the initial solid fraction) differs, no particular attention should be paid to the transient regime. For the highest fluidization velocity, the simulation time was extended in order to achieve statistical convergence. It can be observed that the higher the fluidization velocity, the larger the oscillations of the bed height. It can also be noted that the curves for $U_F \leq 0.090 \text{ m s}^{-1}$ overlap with those of Ozel et al. (2017). Beyond this range, a deviation appears and increases with the fluidization velocity. The bed height is almost always higher than that reported in the simulations of Ozel et al. (2017) for $U_F \geq 0.120 \text{ m s}^{-1}$. It should be noted that the simulations of Ozel et al. (2017) were performed with the same spatial and temporal resolution ($D_p/\Delta x = 12$, $\Delta t_{simu} = 5 \cdot 10^{-4} \text{ s}$). Three major differences should be highlighted between the simulations of this study and those of Ozel: (i) the size of the cross-section, (ii) the geometry of the cross-section, and (iii) the treatment of fluid–solid interfacial friction. Each difference is detailed below. The cross-section in our study is 3.1% larger than in the simulations of Ozel et al. (2017), but this cannot explain the discrepancy, since the simulated bed height should then

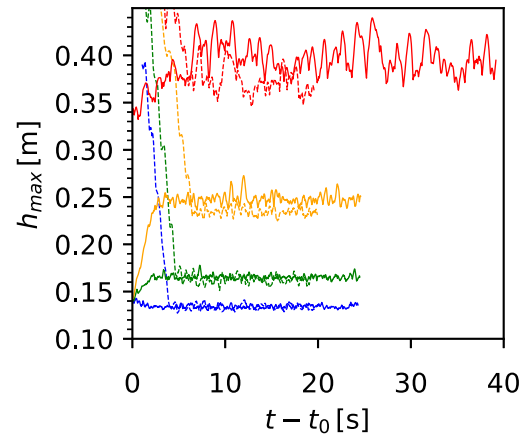


Fig. 2. Temporal evolution of the bed height for various fluidization velocities. — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$. — : numerical work of Ozel et al. (2017), — : present work. $\frac{D_p}{\Delta x} = 12$.

have been lower than in Ozel's simulations. Moreover, it is unlikely that such an effect would only arise at higher fluidization velocities. In their study, the authors used a cylindrical domain (through a Darcy penalization method Khadra et al., 2000). The shape of the cross-section is therefore a better candidate to account for the differences between the curves. Finally, the treatment of interfacial friction also differs. Ozel et al. (2017) defined an apparent particle diameter lower than the real particle diameter. Indeed, with a harmonic viscosity model and at coarse resolution, friction tends to be overestimated if the particle diameter seen by the fluid solver is equal to the real particle diameter (Vincent et al., 2014). However, in the present work, the real diameter was used. This choice was made because a mesh sensitivity analysis is carried out, and the overestimation of friction is assumed to decrease as the grid resolution increases. For the present mesh, with a grid resolution of twelve meshes per particle diameter, it is likely that friction is overestimated. This would also explain why the discrepancy increases with the fluidization velocity. The detailed explanation of these differences is provided in the following paragraphs.

3.2.2. Solid fraction

Several approaches exist to compute the solid fraction. Either from the Lagrangian positions of the particles (see Fig. 2), or using the Eulerian field of the phase indicator function. In this section, the presence rate is computed from the Lagrangian positions of the particles. Then, the temporal average of the evolution of the solid fraction over the bed height is computed (see Fig. 3). In Fig. 3, one can note that the average solid fraction of the data from Ozel et al. (2017) fluctuates more than that calculated from the present work because of less converged statistics. It can be noted that the solid fraction computed in Ozel et al. (2017) is always higher than the one computed in this work. Moreover, the difference does not seem to depend on the fluidization velocity, contrary to what was observed for the bed height. This figure also shows that the temporal average of the particle solid fraction is uniform across the bed height. It should be noted that the solid fraction shows a slight upward trend across the bed height for the highest fluidization velocity, which is also observed in the results of Ozel et al. (2017) and Hamidi et al. (2023), although to a lesser degree. The temporal evolution of the solid fraction (see Fig. 4) in the bed confirms that the regime is established. Finally, these two figures show that all the studied configurations represent a homogeneous fluidization regime.

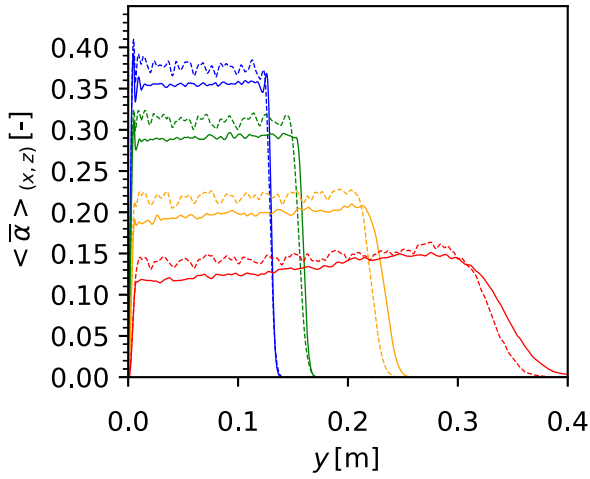


Fig. 3. Time averaged solid fraction as a function of the bed height. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. - - : numerical work of Ozel et al. (2017), — : present work. $\frac{D_p}{\Delta x} = 12$.

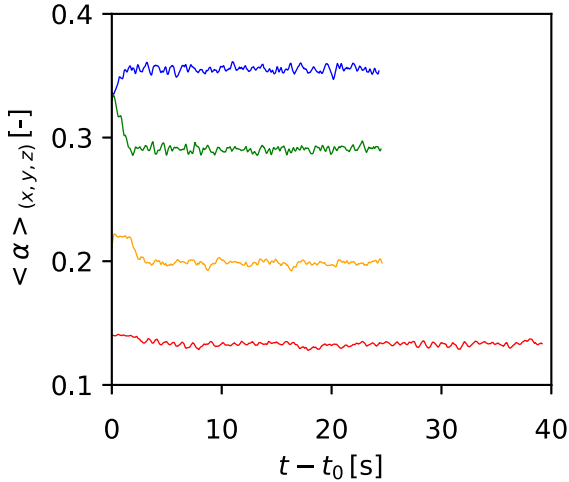


Fig. 4. Temporal evolution of the spatial average solid fraction. The solid fraction is averaged on the volume included between 5% and 85% of the bed height. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. $\frac{D_p}{\Delta x} = 12$.

3.2.3. Influence of the viscosity model

To understand the extent to which the choice of viscosity model affects the physics of the flow, the bed height (see Fig. 5) and the average solid fraction (see Fig. 6) obtained in our simulations are compared with those obtained in the work of Hamidi et al. (2023). The comparison between the simulations performed in this study and those performed by Hamidi et al. (2023) allows to evaluate the influence of the viscosity model, with all other parameters being identical. The viscosity model used by the authors for their study is referred to as the *on-off* viscosity model which is defined as follows:

$$\mu = \begin{cases} \mu_p & \text{if } I = 0 \\ \mu_f & \text{otherwise} \end{cases} \quad (13)$$

It can be seen in Fig. 6 that with the *on-off* model used by Hamidi et al. (2023), the bed height is always lower than that obtained with a harmonic model. Thus, fluid-particle friction is less significant in this case because the *on-off* model reduces the surface area of the particle seen by the fluid solver. This translates into higher values for

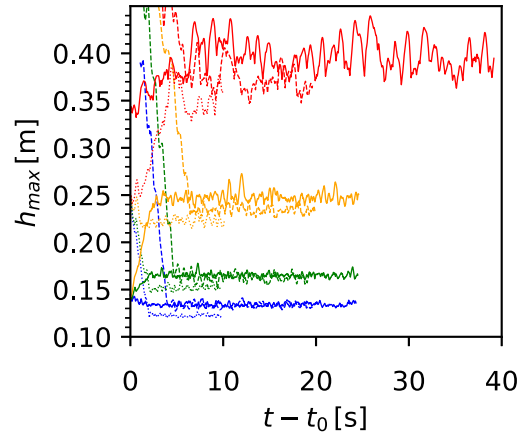


Fig. 5. Temporal evolution of the bed height for different fluidization velocities. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. - - : données de (Ozel et al., 2017), ··· : numerical work of Hamidi et al. (2023), — : present work.

the particles presence rate (see Fig. 6). The solid fraction computed in this case then overlaps with the simulation results of Ozel et al. (2017) for the two highest fluidization velocities. For the two lowest fluidization velocities, the solid fraction computed by Hamidi et al. (2023) is slightly higher than that calculated by Ozel et al. (2017), but the difference is less than 4% on average. Compared to the experimental results (see Fig. 7), the harmonic viscosity model shifts the curve slightly to the left compared to the *on-off* model of Hamidi et al. (2023). This phenomenon was expected and has been previously described in Vincent et al. (2014). Indeed, using the *on-off* viscosity model is equivalent to reducing the particle diameter used to compute the friction, as shown in Vincent et al. (2014). In the present work, the harmonic viscosity model was employed to quantify the reduction in friction error with increasing resolution. However, the current grid resolution is not sufficient to correct the overestimation of friction (see Section 3.3). Therefore, the interested reader is advised to use the *on-off* viscosity model or to adopt a reduced particle diameter equal to $D_p - \frac{\Delta x}{2}$ to compute the friction, as recommended in Vincent et al. (2014). If the boundary layer is fully resolved, the use of the harmonic viscosity model is recommended.

3.2.4. Fluidization law

Fig. 7 provides information on the fluidization law. It represents the fluidization velocity as a function of the average solid fraction in the bed. The latter is computed by taking the time average of the space average solid fraction of the particles between 5% h_{max} and 85% h_{max} . The computed solid fraction is consistent with the numerical and experimental results in the literature, indicating that the Richardson-Zaki fluidization law (Richardson and Zaki, 1954) is correctly predicted. The slight shift to the left of our results compared to the Richardson-Zaki curve can be explained by the choice of viscosity model, as explained above.

3.3. Effect of the grid resolution on the solid fraction

Three grid resolutions were studied: 12, 24, and 36 meshes per particle diameter. Due to the numerical cost of the simulations, the finest resolution was only studied for the lowest fluidization velocity. In addition, the simulated physical time associated with resolutions of 24 meshes per diameter is reduced for certain cases. The computation times are reported in Table 2. Priority in terms of numerical cost was given to the two velocities defining the boundaries of the study.

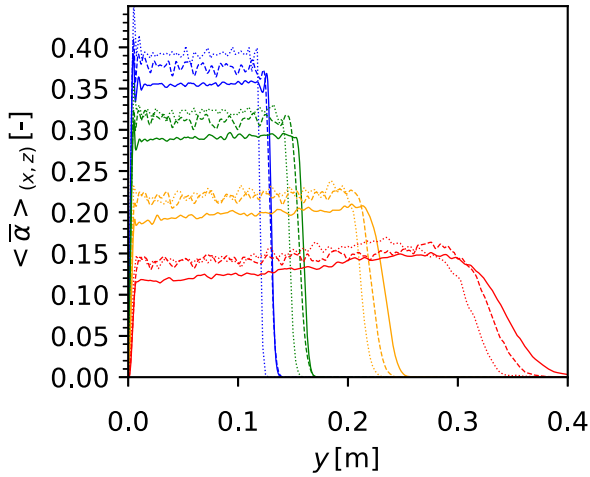


Fig. 6. Evolution of the void fraction along the bed height — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$. — : numerical data of Ozel et al. (2017), ··· : numerical work of Hamidi et al. (2023), — : present work.

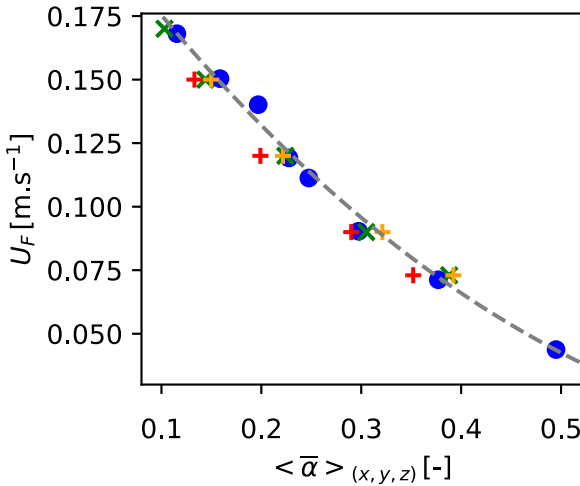


Fig. 7. Fluidization law. —: Richardson-Zaki law (Richardson and Zaki, 1954), ·: experimental work of Corona (2008), ×: numerical work of Ozel et al. (2017), +: numerical work of Hamidi et al. (2023), +: present work. $\frac{D_p}{\Delta x} = 12$.

Table 2

Simulated physical time (in seconds) for each configuration.

$D_p/\Delta x$	$U_F [\text{m s}^{-1}]$	0.073	0.090	0.120	0.150
12		25	25	25	39
24		23	6.6	4.9	9.7
36		10	—	—	—

Fig. 8(a) highlights that, across all configurations studied, few fluctuations are observed in the temporal evolution of the average particles presence rate. This allows representative averages to be computed over relatively short physical times. In addition, the grid resolution appears to have negligible influence on this quantity. The curves are superimposed. Thus, even with a coarse mesh (about ten meshes per particle diameter), the bed compacity is correctly predicted. This was expected, as the fluidization law is correctly predicted with a coarse mesh (see Fig. 7). More differences are observed in the maximum bed height as shown in Fig. 8(b). For a fluidization velocity of 0.15 m s^{-1} ,

the height of the highest particle in the bed is lower for the resolution of 24 meshes per diameter than for that of 12 meshes per diameter, and the fluctuations are smaller. For a fluidization velocity of 0.073 m s^{-1} , the bed height increases slightly with the grid resolution, but the difference is negligible compared to the bed height. The solid fraction as a function of vertical position is shown in Fig. 9. The fluctuations associated with the finest grid resolutions are slightly higher than for the coarsest resolution. It can be attributed to the shorter computation times, which prevent full statistical convergence. However, the curves overlap, indicating that the coarse resolution is also suitable for studying this quantity. A remarkable result is observed for the highest fluidization velocity (shown in red in Fig. 9). The presence rate appears to be more homogeneous across the bed height than for the coarse resolution. This slight increase in particles presence with height had been observed in Ozel et al. (2017) and Hamidi et al. (2023), where statistics were computed over 10 s and 5 s, respectively. Since the average was computed over 10 s for the resolution of 24 meshes per diameter and over 22 s for the resolution of 12 meshes per diameter, the results are comparable. This increase in the presence rate as a function of height does not seem to be an effect of lack of time convergence, but rather an effect of spatial under-resolution. This effect also seems to be influenced by the choice of viscosity model, as shown in Fig. 6. Thus, this phenomenon is influenced by both the choice of viscosity model and the grid resolution, indicating under-resolution of friction at the fluid-particle interfaces. It is expected that a resolution exceeding 48 cells per particle diameter would be required to correct the overestimation of friction discussed in Section 3.2.3, even at the lowest fluidization velocity. This would lead to a slight rightward shift of the computed solid fraction to align with the experimental fluidization law (see Fig. 7).

The study of the presence rate, bed height and fluidization law showed that the macroscopic behavior of the fluidized bed is well predicted. Slight differences appear between numerical and experimental results, which can be explained by the different cross-section area and the choice of the viscosity model. Finally, grid resolution does not have a significant influence of these quantities. In the following, a more in-depth analysis is performed on Eulerian and Lagrangian quantities in order to describe the agitation of the fluid-particle flow.

4. Lagrangian statistics of the particles agitation

Lagrangian statistics are computed with the only knowledge of the position and the velocity of the particles' gravity center. First, the trajectories are analyzed in Section 4.1. The spatial distribution of particles described with the radial distribution function of particles pair is reported in Section 4.2. The velocity and the velocity variances of the particles' gravity center are detailed in Sections 4.3 and 4.4, respectively.

For hereinafter, the notation $\bar{\cdot}$ denotes for the time average, the notation $\langle \cdot \rangle_{i,j}$ for the spatial average in direction i , and j , and the notation $\langle \cdot \rangle_p$ the ensemble average over the particles. For more details on the way statistics are computed, the reader is referred to Appendix A of Butaye et al. (2025).

$$G(r) = \frac{\frac{4}{3}\pi R_p^3 N_p(r)}{\langle \bar{\alpha} \rangle V_c(r)} = \frac{R_p^3 \sum_k \sum_{i=1}^{N_p} \sum_{j=1}^{N_p} (H_{ij}(r_{ij}(k\Delta t) - r + \Delta r) - H_{ij}(r_{ij}(k\Delta t) - r - \Delta r))}{((r + \Delta r)^3 - (r - \Delta r)^3) N_p N_t \Delta t \langle \bar{\alpha} \rangle} \quad (14)$$

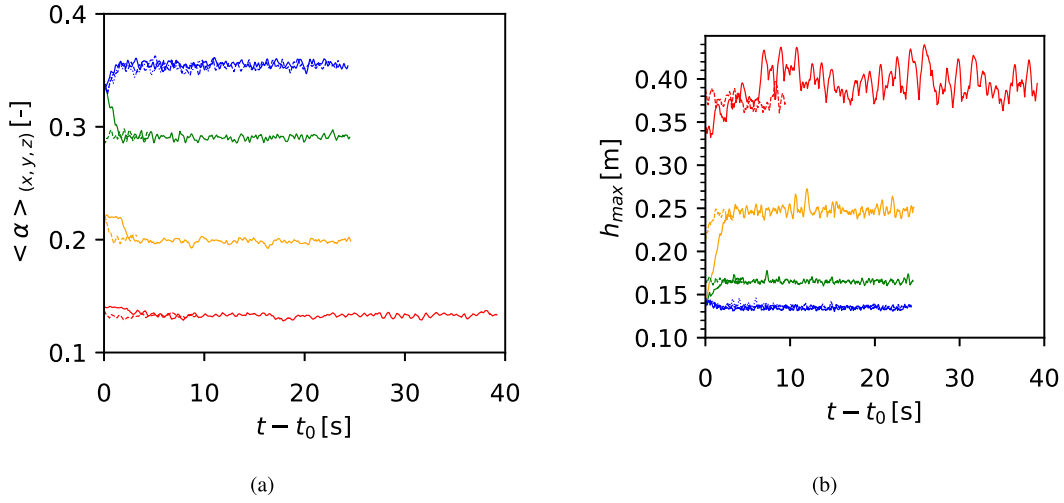


Fig. 8. Temporal evolution of: (a) the solid fraction, (b) the bed height. — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$. — : $\frac{D_p}{\Delta x} = 12$, — : $\frac{D_p}{\Delta x} = 24$, ... : $\frac{D_p}{\Delta x} = 36$.

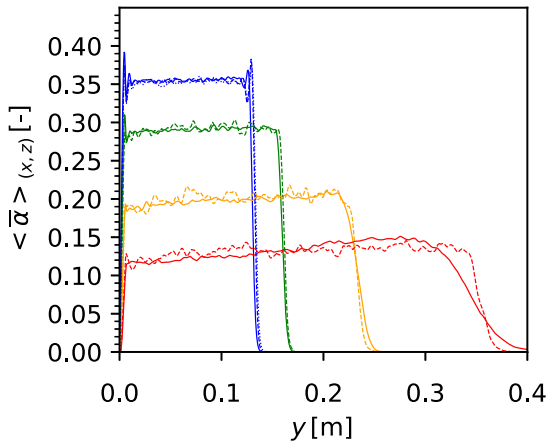


Fig. 9. Time averaged evolution of the solid fraction as a function of the bed height. — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$. — : $\frac{D_p}{\Delta x} = 12$, — : $\frac{D_p}{\Delta x} = 24$, ... : $\frac{D_p}{\Delta x} = 36$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

4.1. Particle trajectories

Lagrangian tracking of each particle provides access to its trajectory. Analysis of trajectories can reveal both large-scale physical phenomena, which are characterized by long timescales (low frequencies), and small-scale physical phenomena, which are characterized by short timescales (high frequencies) (Corona, 2008). The projection of the trajectories of eight particles in the horizontal and vertical planes is shown in Fig. 10. It can be seen that the higher the solid fraction, the less the particles travel a significant portion of the domain. It can also be seen that the particles are more often deflected from their trajectories due to collisions and that their movement can be assimilated to Brownian motion. Furthermore, no preferential path seems to be favored by the particles, which is consistent with the homogeneous fluidization regime. Corona (2008) observed a toroidal motion in which particles tend to rise in the center of the domain and fall toward the walls. This motion is difficult to identified on this figure but will be discussed in Fig. 14.

The temporal evolution of the height and the vertical velocity of a particle initially located at mid-height in the domain is shown in

Fig. 11 for the cases studied. The oscillations of the horizontal velocity component are slightly smaller for the highest solid fraction, but are of the same order of magnitude in the other cases. This is contrary to what is observed in Corona (2008), where the fluctuations of the vertical component are slightly larger in all configurations. The velocity fluctuations reveal both small-scale (high-frequency) motion — probably due to particle collisions and fluid agitation — and large-scale (low-frequency) motion. The norm of the velocity fluctuations increases with the fluidization velocity, but the same behavior is observed, reflecting the similarity of the physical phenomena governing the flow in the four configurations.

4.2. Spatial distribution of particles

A more localized measure of the spatial distribution of particles than the solid fraction can be obtained by computing the average distance to the nearest particle. This quantity provides a more precise insight into the interstitial space between particles. It is computed by considering all particles, including those in collision, for which this distance is zero. As shown in Fig. 12(a), the average distance increases with the fluidization velocity, rising from 1.3% of D_p for $U_F = 0.073 \text{ m s}^{-1}$ to 12% for $U_F = 0.150 \text{ m s}^{-1}$. The figure also reveals a large-scale motion whose frequency decreases as the fluidization velocity increases. The average particle distribution along one of the horizontal directions is presented in Fig. 12(b). The averages are computed over vertical slices of height h_{max} , thickness $D_p/4$, and depth equal to the domain width. The number of particles remains constant for $-4.5 \leq x/D_p \leq 4.5$, then oscillates near the walls, reaching a maximum value at $x/D_p = \pm 5.375$, which corresponds to a distance slightly greater than one particle radius from the wall.

Another method for studying the homogeneity of particle distribution in the bed is to consider the radial distribution function of particle pairs $G(r)$ (Bouche et al., 2012; Alm  ras et al., 2021). $G(r)$ gives an indication of the average presence rate at a distance r relative to the average presence rate in the bed. If $G(r) > 1$, this indicates a significant particle clustering, whereas $G(r) < 1$ reflects a particle depletion in the corresponding region. The distribution function is computed by 14. In this equation, $\langle \bar{\alpha} \rangle$ is the spatio-temporal average of the particle solid fraction computed between 5% h_{max} and 85% h_{max} , $V_c(r)$ is the control volume of the shell located at r and with thickness Δr , $N_p(r)$ is the number of particle gravity centers between $r - \Delta r$ and $r + \Delta r$. For the analysis below, we set $\Delta r = 0.1 D_p$. $H(x)$ is the Heaviside function, equal to 0 if $x < 0$ and 1 otherwise. Δt is the simulation time step and N_t is the

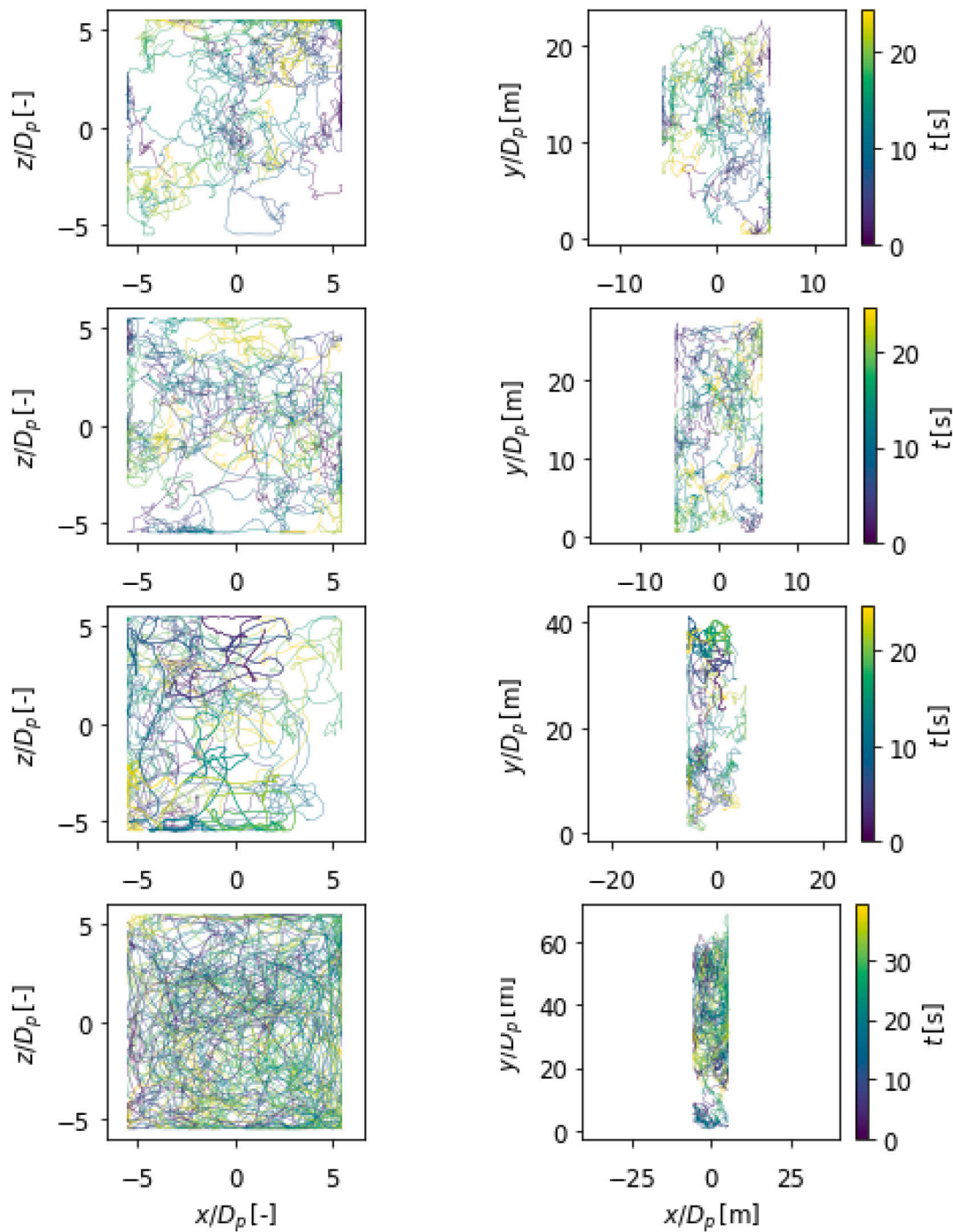


Fig. 10. Projection of particle trajectories in the horizontal plane (left) and in the vertical plane (right). The color scale corresponds to the time evolution. $\frac{D_p}{\Delta x} = 12$. From top to bottom: $U_F = 0,073 \text{ m s}^{-1}$ ($\alpha = 0,35$), $U_F = 0,090 \text{ m s}^{-1}$ ($\alpha = 0,29$), $U_F = 0,120 \text{ m s}^{-1}$ ($\alpha = 0,20$), $U_F = 0,150 \text{ m s}^{-1}$ ($\alpha = 0,13$). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

number of time steps considered. r_{ij} represents the distance between the centers of gravity of particles i and j .

The curves are presented in Fig. 13 for the four configurations studied. The function is computed in two different ways. First, by considering all particles located between 5% and 85% of the bed height (see Fig. 13(a)). As observed by Alm  ras et al. (2021), there is a significant agglomeration of particles, up to four times the average concentration, in the immediate vicinity of the particles. It should be noted that $N(r)$ corresponds to the number of particle gravity centers between $r - \Delta r$ and $r + \Delta r$, and not to the precise computation of the local

solid fraction. Although this pseudo-presence rate is not physical, as it is greater than the maximum compactness of a fixed bed (and greater than 1 for the lowest fluidization velocities), it provides information on how the solid fraction evolves as a function of the distance to a given particle. Thus, the higher the solid fraction, the greater the accumulation of particles at short distances.

A peak is observed at $\frac{r}{D_p} \sim 1$ which is explained by physical and geometrical considerations as it corresponds to the distance of collision between two particles. The same peak was observed in Alm  ras et al. (2021), though less pronounced, reaching a value of 1.6 for a

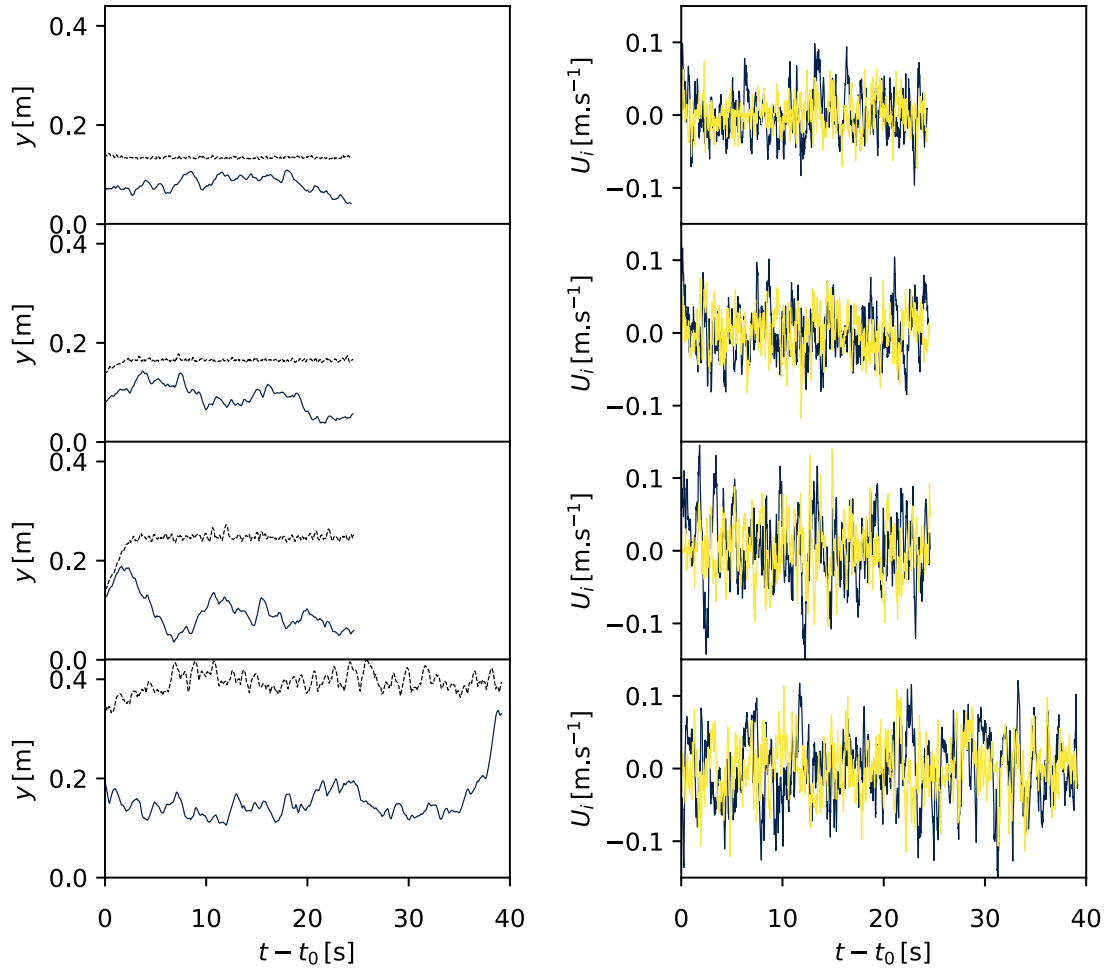


Fig. 11. Temporal evolution of the vertical position (left), and the velocity (right) of a particle initially at the mid-height of the bed for the grid resolution $\frac{D_p}{\Delta x} = 12$. From top to bottom: $U_F = 0.073 \text{ m s}^{-1}$, $U_F = 0.090 \text{ m s}^{-1}$, $U_F = 0.120 \text{ m s}^{-1}$, $U_F = 0.150 \text{ m s}^{-1}$. ■ : U_x , ■ : U_y , - - : h_{max} .

solid fraction of 0.42. Their function was obtained experimentally by measuring the distances between pairs of particles passing through a laser sheet. Consequently, their evaluation is based on particles within a two-dimensional section, in contrast to the method used in this work, where particles are counted within a spherical shell surrounding a given particle. As a result, more particles can be identified at short distances, which explains the higher peak values of the distribution function in the present approach. A second peak is observed at a distance of two particle diameters, with a value close to 1. This peak tends to vanish as the solid fraction decreases, consistent with the findings of [Almérás et al. \(2021\)](#). The distribution function then decreases again and does not tend toward unity, indicating a non-homogeneous distribution beyond two particle diameters. This behavior reflects a confinement effect imposed by the finite size of the computational domain. Indeed, the pair distribution function is anisotropic near the walls. To verify that the distribution is homogeneous in the core region of the bed, the function was computed by considering only particles located at least four diameters away from the walls (see [Fig. 13\(b\)](#)). In this case, the distribution is close to 1 for distances between two and four particle diameters, indicating a homogeneous distribution within the bed. The decrease of the curve beyond four diameters is attributed to the lateral confinement by the walls. The right-hand figure also shows results obtained with a grid resolution of 24 meshes per particle diameter. Only minor differences are observed between the two grid resolutions, which is consistent with the mean bed height (see [Fig. 8\(b\)](#)) and the average solid fraction (see [Fig. 8\(a\)](#)).

4.3. Particles' velocity

The study of particle velocity makes it possible to characterize the agitation of the fluidized bed. First, the profile of the vertical velocity in the cross-section area are analyzed. Then, probability density functions of the fluctuating velocity are investigated.

4.3.1. Profile of the vertical velocity in the cross-section area

In the configuration under study, the fluidized bed is not transported, which results in a zero mean particle velocity. Examining the mean velocity profile as a function of the transverse direction makes it possible to determine whether preferential zones exist for rising and descending particles. To do this, particle velocities can be averaged within vertical slices of the domain, following the same approach as for computing the mean profile of the particle number (see [Fig. 12\(b\)](#)). The mean profiles of the particle vertical velocity for each fluidization velocity are shown in [Fig. 14](#). In this figure, each cross corresponds to the average computed over a vertical slice of the flow with a thickness of $D_p/4$. On the plot, the crosses are located at the middle of the slice. It can be observed that particles tend to rise in the center of the domain and to descend near the walls. The descending region near the wall can reach a thickness of up to four particle diameters. In a one diameter thick region close to the walls, the velocity profile decreases before steadily increasing toward the center of the domain. Because of the square cross-section, fluidization is not homogeneous near the walls. The behavior of particles near the wall will therefore not be considered in the computation of Eulerian statistics. The orange profile,

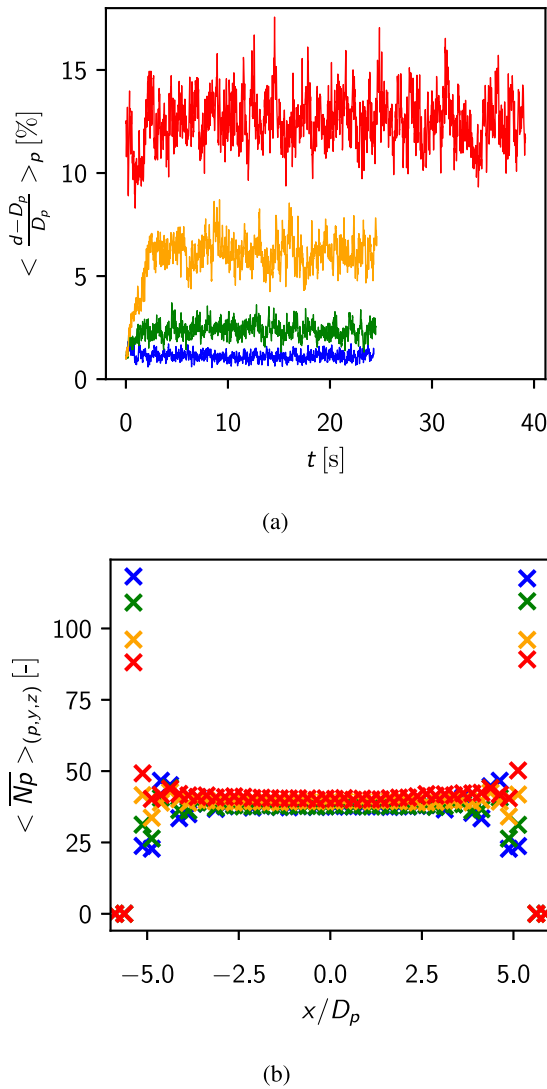


Fig. 12. (a) Average distance to the gravity center of the nearest particle. (b) Average particle number profile as a function of position in the horizontal direction. The averages are computed on slices of width $D_p/4$ over the entire height of the bed. — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$.

associated with a fluidization velocity of 0.12 m s^{-1} , is asymmetric and shows high particle upward velocities in the vicinity of the wall, which can be explained by insufficient temporal convergence of the statistical quantities. The time convergence study and grid sensitivity analysis are presented in [Appendix A](#). A more continuous representation of the vertical particle velocity is examined using Eulerian fields in [Section 5.1](#).

4.3.2. Velocity probability density functions

The Probability Density Functions (PDFs) of particles' velocity are widely studied in the literature to characterize fluidized bed dynamics ([Risso, 2016](#); [Corona, 2008](#); [Alm  ras et al., 2021](#)). The PDFs of the vertical and horizontal components of the Lagrangian particle velocity are shown in [Fig. 15](#) for the four configurations considered. In this figure, as well as in [Fig. 16](#), the PDFs are normalized such that the integral of each function equals 1. In [Fig. 15](#), a rightward shift of the PDFs of the vertical velocity relative to a Gaussian distribution is observed. This is consistent with the explanations of [Amoura et al. \(2017\)](#), who attributes this phenomenon to spatial fluctuations, i.e., to

the mean disturbance generated in the vicinity of each sphere. It can also be noted that, once non-dimensionalized by the standard deviation, the PDFs are independent of the fluidization velocity and therefore of the particle Reynolds number. The slight offset between curves is explained by insufficient time convergence in some simulations. For the cases where convergence is achieved, corresponding to the lowest fluidization velocity (blue curve) and the highest fluidization velocity (red curve), the curves are superimposed.

The rightward shift is more pronounced than what was observed by [Alm  ras et al. \(2021\)](#), where the studied configuration is similar to the present work. Regarding the horizontal velocity component, the PDFs are symmetric and follow a Gaussian distribution, except at the origin where a peak is observed. This peak does not appear to depend on the fluidization velocity, unlike what was reported in [Alm  ras et al. \(2021\)](#), where the peak amplitude increases with the solid fraction.

To compare with the experimental results of [Corona \(2008\)](#), [Fig. 16](#) shows the PDFs of the non-dimensionalized fluctuating velocity. The rightward shift of the vertical velocity PDFs is again observed, as well as the amplitude peak at the origin. However, in [Figure 3.12 of Corona \(2008\)](#), this peak amplitude relative to the associated Gaussian increases with solid fraction, which is not observed in the present case. On the other hand, as in Corona's work, the standard deviation decreases with increasing solid fraction, implying a narrower velocity distribution and therefore a larger peak amplitude at the origin. The numerical values associated with the PDFs differ from those of the author due to what appears to be a different normalization: in particular, the integral of the author's PDFs is not equal to 1.

Regarding the horizontal component, the PDFs are symmetric and perfectly Gaussian with a peak at the origin. [Corona \(2008\)](#) attributes this peak to inelastic collisions. When the impact velocity is too low, resulting in a collision Stokes number less than or equal to 10, the restitution coefficient becomes zero and the particles do not rebound, but instead remain in contact.

4.4. Particle velocity variances

The variance of the vertical and horizontal components of the particle velocity is shown in [Fig. 17](#). For each component, the variance of the fluctuating velocity decreases with increasing concentration. The discrepancies observed between the various numerical results can be explained both by differences in the viscosity models used and by insufficient time convergence. The latter aspect is examined in [Appendix B](#). The velocity variance is well predicted for a solid particle volume fraction higher than 20%. For the highest fluidization velocity, the discrepancies between the experiments and the numerical simulations exceed 50%. One possible explanation is the way in which the variances were computed. Experimentally, the author used data measured on 12 tracer particles over 204 s. Numerically, the averages were computed on the 2134 particles of the bed, over approximately ten seconds for [Ozel et al. \(2017\)](#) and [Hamidi et al. \(2023\)](#), and between twenty and thirty-two seconds in the present work. [Corona \(2008\)](#) performed a convergence study on the number of particles used for averaging. The acquisition time associated with the chosen number of particles ensured converged data. In this study, time convergence is not reached for the highest fluidization velocity (see [Appendix B](#)), but the trend does not suggest convergence toward the experimental data. Therefore, the difference in computations method cannot explain these discrepancies. One should also note that the present results are very close to the one obtained by the numerical study of [Ozel et al. \(2017\)](#).

In [Figure 4 of Alm  ras et al. \(2019\)](#), who experimentally studied a fluidized bed similar to that of [Corona \(2008\)](#), it is observed that the variance of the vertical fluctuating velocity of the solid particles increases as the solid fraction decreases. The change in slope seen in [Fig. 17](#) does not appear in their case. Thus, the validity of the experimental measurements of [Corona \(2008\)](#) at the highest fluidization velocities may be questioned due to potentially unconverged statistics.

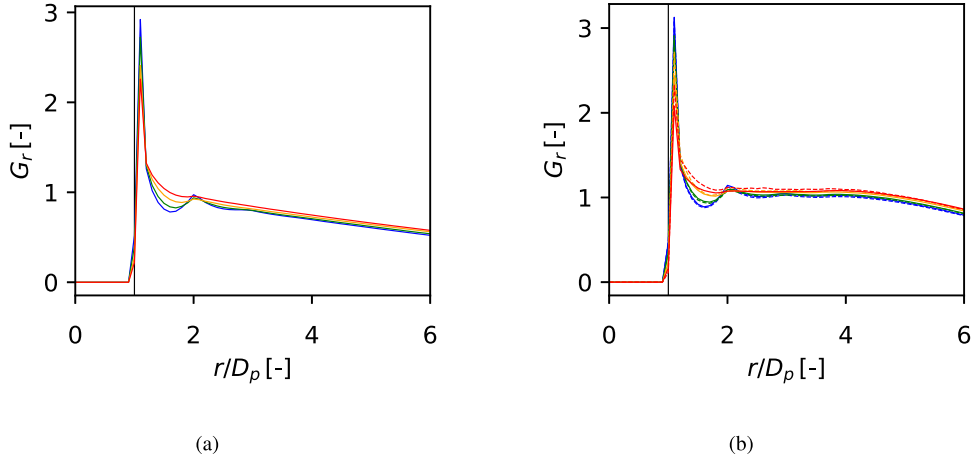


Fig. 13. Radial pair distribution function. The function is computed considering only particles between 5% and 85% of the bed height. The black vertical line corresponds to the x-coordinate $\frac{r}{D_p} = 1$. (a) all particles between 5% and 85% of bed height are considered. (b) only particles at a distance greater than or equal to four particle diameters from the walls are considered. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. — : $\frac{D_p}{D_x} = 12$. - - : $\frac{D_p}{D_x} = 24$.

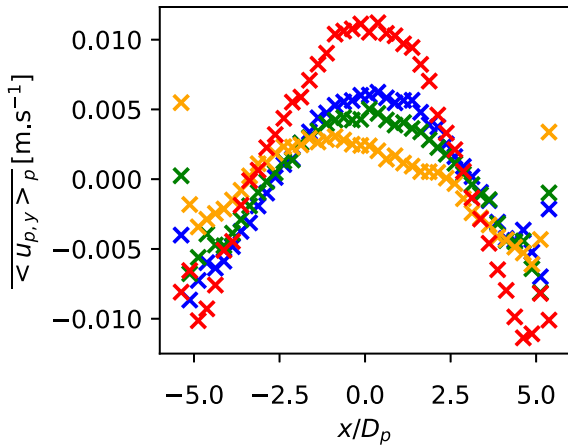


Fig. 14. Vertical velocity profile of solid particles computed in a Lagrangian manner. $\frac{D_p}{D_x} = 12$. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$.

The anisotropy coefficient is always greater than 1 and remains constant for $\alpha > 0.2$ (see Fig. 18). The numerical results underestimate the anisotropy coefficient compared to the experimental results. However, the trend over the range of solid fractions studied is correctly predicted. The fact that this coefficient is constant suggests that the same mechanisms govern the flow for the four fluidization velocities considered. Moreover, as emphasized by Ozel et al. (2017), the large-scale motion of the particles is correctly predicted.

4.5. Characteristic time scale of particle diffusion

The analysis of the random motion of particles is essential for understanding the flow and for multiscale modeling approaches. Since particle motion in a fluidized bed is chaotic, particle velocity fluctuations become decorrelated after a certain amount of time (Abbas, 2024; Corona, 2008). A resulting induced diffusion can then be defined by analogy with a particle undergoing Brownian motion, even though the particles are not Brownian. This diffusion is induced both by fluid-particle interactions and by particle-particle collisions (Corona, 2008). The decorrelation time corresponds to the characteristic time over which fluctuating velocity is diffused within the bed. In this sense, a

characteristic timescale can be determined in the vertical direction and another in the horizontal direction.

The diffusion coefficient, and consequently the Lagrangian characteristic time, can be determined in two ways, as shown by Corona (2008). By expressing the displacement variance as a function of the integral of velocity, the author established the following relation:

$$\overline{\langle (x_i - x_{i,0})^2 \rangle_p} = \left\langle \lim_{t \rightarrow \infty} \left(2 \overline{\langle u_{p,i}^2 \rangle_p (t - t_0)} \int_{t_0}^{\infty} R_{ii}(\tau) d\tau \right) \right\rangle_{t_0} \quad (15)$$

where the subscript $_0$ refers to a given initial time after the statistically steady regime has been reached. In the present study, the averages are taken once the bed has reached its mean height. One also note:

$$R_{ii}(\tau) = \frac{\langle u_i(t_0) u_i(t_0 + \tau) \rangle_{p,t_0}}{\langle u_{p,i}^2(t_0) \rangle_p} \quad (16)$$

$$t_{L,ii} = \int_0^{\infty} R_{ii}(\tau) d\tau \quad (17)$$

$$M_{ii}(t - t_0) = \langle (x_i(t) - x_i(t_0))^2 \rangle_p \quad (18)$$

$$D_{ii} = \lim_{t \rightarrow \infty} \left(\frac{1}{2} \frac{d}{dt} \langle (x_i(t) - x_i(t_0))^2 \rangle_p \right) = \overline{\langle u_{p,i}^2 \rangle_p} t_{L,ii} \quad (19)$$

where D_{ii} is the self-diffusion coefficient. Corona proposed determining D_{ii} by graphical evaluation, taking the limit at infinity of the curve $\frac{M_{ii}}{2(t-t_0)}$. R_{ii} is the autocorrelation coefficient of the (i)th component of the particle velocity. Finally, $t_{L,ii}$ is the Lagrangian timescale associated with particle diffusion.

This method makes it possible to evaluate the self-diffusion coefficient in two different ways and thus to validate the results obtained. Corona (2008) reported an average discrepancy between the two values of about 20%, with a maximum error of 35% in the horizontal direction and 23% in the vertical direction. Furthermore, $t_{L,ii}$ is systematically compared with the characteristic timescales of the flow, namely: (i) the particle response time due to viscous drag, (ii) the mean time between two collisions, (iii) the mean fluid motion time at the particle scale, and (iv) the characteristic timescale of the fluid fluctuations. This comparison makes it possible to identify the scales at which vertical and horizontal diffusion occur and to relate them to characteristic length scales of the flow. In this paper, velocity autocorrelation and particle self-diffusion are examined in order to extract the characteristic temporal and spatial scales of particle diffusion induced by the flow.

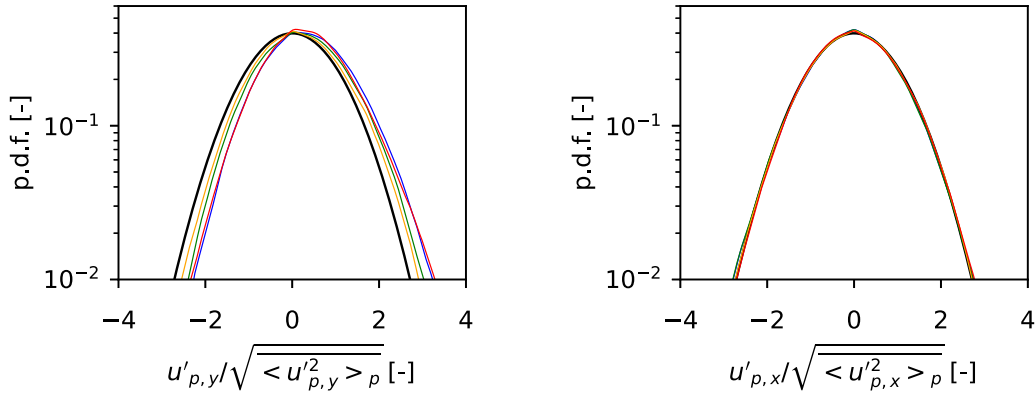


Fig. 15. Probability density function of particle velocity fluctuations normalized by their standard deviation. Vertical component (left) and horizontal component (right). — : gaussian curve. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

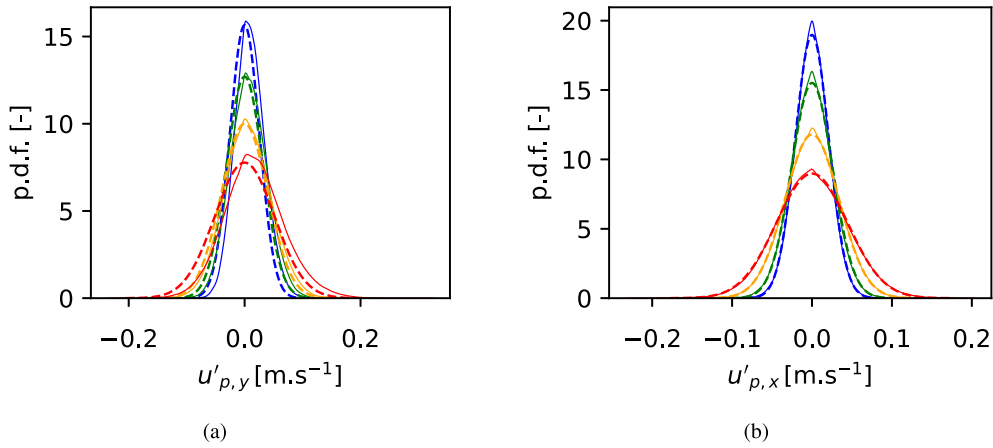


Fig. 16. Probability density function of particle velocity fluctuations. Vertical component (left) and horizontal component (right). — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. — : simulation result. - - : gaussian curve with the same standard deviation as the associated simulation.

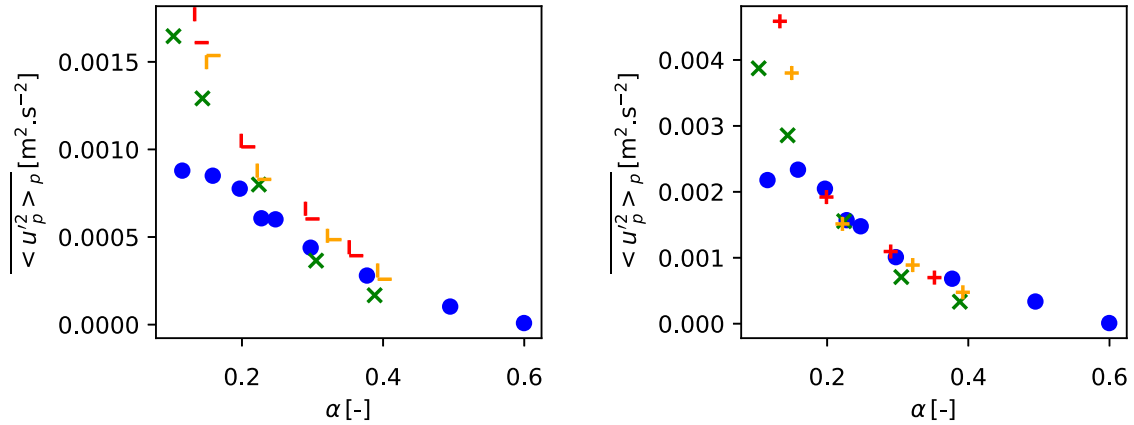


Fig. 17. Variance of horizontal (left) and vertical (right) components of particle velocity for $\frac{D_p}{\Delta x} = 12$. • : experimental work of Corona (2008). × : numerical work of Ozel et al. (2017), + : numerical work of Hamidi et al. (2023), × : present work.

4.5.1. Autocorrelation of particle velocities

The autocorrelation coefficients of particle velocities were computed by considering all the particles in the domain and an initial time t_0 every 10^3 simulation time steps. The coefficients are shown in Fig. 19 for resolutions of 12 and 24 meshes per particle diameter for the four fluidization velocities, as well as for a resolution of 36 cells per particle diameter for the lowest fluidization velocity. The convergence

of the ensemble average of the autocorrelation coefficients for the two resolutions is presented in Appendix C.

In the left-hand figure, for $\frac{D_p}{\Delta x} = 12$, it is observed that the autocorrelation of the horizontal velocity decreases toward 0 more rapidly than that of the vertical velocity. In both cases, the functions oscillate around zero. The amplitude of these oscillations decreases with increasing solid fraction. Except for the vertical component of the lowest fluidization

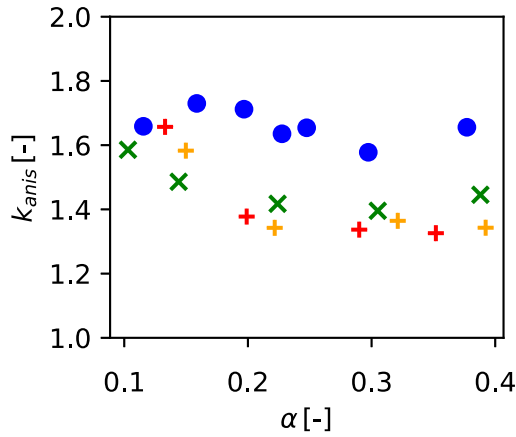


Fig. 18. Anisotropy coefficient of velocity fluctuations for $\frac{D_p}{\Delta x} = 12$. • : experimental work of Corona (2008). × : numerical work of Ozel et al. (2017), + : numerical work of Hamidi et al. (2023), + : present work.

velocity, the curves indicate an anticorrelated motion when passing through negative values. Velocity fluctuations become fully decorrelated after a maximum time of 4 s for the highest fluidization velocity. Similar oscillations were observed in the numerical study of Ozel et al. (2017), who attributed these to recirculations in the lower part of the bed. These oscillations were not observed in the experimental curves, as these were obtained with only 12 particles.

In the right-hand figure, for $\frac{D_p}{\Delta x} = 24$, a completely different behavior is observed. The autocorrelation associated with vertical velocity fluctuations decreases much more slowly toward 0 than for the coarser resolution, and the curves show no anticorrelated motion. The curves drop toward 0 more rapidly as the volume fraction increases. The oscillations observed at 12 cells per particle diameter therefore appear to be a numerical effect related to under-resolution of the flow.

The characteristic time $t_{L,ii}$, computed by integrating the curves from Fig. 19, is shown in Fig. 20. We denote $t_{L,v} = t_{L,yy}$ as the characteristic diffusion time in the vertical direction and $t_{L,h} = (t_{L,xx} + t_{L,zz})/2$ as the characteristic diffusion time in the horizontal direction. According to experimental data, $t_{L,v}$ decreases for a solid fraction between 0.1 and 0.2, then increases with the volume fraction up to 0.5 and appears to decrease again. More experimental points would be required for volume fractions above 0.4 to accurately describe the evolution of $t_{L,ii}$. Moreover, the data for $\alpha = 0.6$ do not appear to be converged. Indeed, in Figure 5.7 of Corona (2008), $t_{L,xx} \neq t_{L,zz}$, which should not be the case, as the flow is isotropic in the horizontal plane.

$t_{L,h}$ increases for volume fractions between 0.1 and 0.4, then remains constant up to 0.6. Regarding the values obtained from numerical simulations for $\frac{D_p}{\Delta x} = 12$ (in blue in the figure), $t_{L,h}$ does not follow the same increase as the experimental values and is underestimated. For the resolution $\frac{D_p}{\Delta x} = 24$ (in orange), the $t_{L,h}$ values are constant and approximately 0.06 s. The influence of resolution on $t_{L,h}$ is more significant at low volume fractions and is even more pronounced for $t_{L,v}$ than for $t_{L,h}$. Indeed, for the lowest volume fraction, the $t_{L,v}$ value obtained at 24 meshes per particle diameter is 4.6 times larger than at 12 meshes per particle diameter, compared to a ratio of 2.4 for $t_{L,h}$.

For the two lowest solid fractions, the decorrelation time is consistent with experimental values, although slightly lower. In contrast, $t_{L,v}$ remains largely underestimated for the two highest volume fractions. Nevertheless, a change in slope is observed for $\alpha \geq 0.29$. For these values, the grid resolution seems to have little influence on the results, even though time convergence has not been reached and the values may increase slightly (see Appendix C).

One possible explanation for the large discrepancy between numerical and experimental values is the geometry of the section. In

the numerical study, the cross-section area is square, whereas it is circular in the experiment. The presence of corners in this configuration could be one of the causes of the observed discrepancies. In particular, particles' accumulation in the corners has been observed, which could affect the computation of velocity autocorrelation coefficients. This phenomenon is represented in Fig. 22. The local solid fraction in the corners can reach values up to twice the mean solid fraction of the bed. Simulations of the fluidized bed in a cylindrical domain (using a Darcy penalization method as done in Ozel et al., 2017) would be required to assess this hypothesis.

4.5.2. Self-diffusion

According to Eq. (20), the self-diffusion coefficient can be computed by examining the variance of the particle trajectories, which is shown in Fig. 21. As with Corona (2008), a quadratic evolution can be observed in the short term, also described as ballistic by Brändle de Motta et al. (2013). This can be explained by analogy with the ballistic trajectory of a projectile subject only to its own weight, whose trajectory depends on time in a quadratic manner. This behavior appears to be well verified here, except for the highest fluidization velocity, where the slope order is greater than 2. Then, for times between 1 s and 10 s, the evolution is linear, which is characteristic of a diffusive regime. By analogy with Brownian motion, the trajectory variance depends linearly on time. The same behavior is observed for time scales of the same order of magnitude in the experimental study by Corona (2008) and in the numerical study of a fluidized bed of 512 particles in a turbulent flow Brändle de Motta et al. (2013). However, this trend does not appear to hold at the highest fluidization velocity. Corona (2008) also observed that the trajectory variance tends toward a constant value in all the three directions. In the present work, the simulation time is too short to observe this stabilization. However, the curves in the horizontal direction draw closer to each other and appear to converge toward the same value. One can also note that the trajectory variance in the vertical direction increases with fluidization velocity. Nevertheless, further simulations would be required to assess these assumptions.

The self-diffusion coefficient can be identified as the converged value obtained in Fig. 23. In the top figure, one can note that the plateau value is constant for the horizontal component at the three lowest velocities, while the curve associated with the highest velocity is not yet converged. The plateau obtained for the vertical components shows an overall increase with the fluidization velocity. However, the curve obtained for the lowest fluidization velocity seems to contradict this result. It should be noted that, unlike the autocorrelation curves, these curves were obtained by considering only a single instant t_0 . Indeed, the temporal convergence of the displacement variance is slower and does not yet appear to have been achieved (see Fig. 21). It is therefore very likely that the overall average between particles is not sufficient to reach convergence. The bottom figure illustrates the influence of resolution on the vertical velocity components. The computation time is too short to reach convergence, but a strong effect of the grid resolution is identified. The values of the coefficients D_{ii} are reported in Table 3 and compared to those obtained from the right-hand side of Eq. (20). The comparison is only possible for the resolution of 12 meshes per particle diameter due to computation time limitation. The coefficients obtained using the displacement variance (DV) are always lower than those obtained using the autocorrelation function (ACF). Moreover, they do not exhibit a strictly increasing trend with solid fraction, unlike the ACF-based values. The influence of the grid resolution on the ACF clearly highlights the under resolution of the flow for the coarser mesh. Finally, for the grid resolution of 24 meshes per particle diameter, a better agreement with experimental results is obtained for the highest fluidization velocities (lowest solid fractions). The discrepancy falls below 6% for $\alpha = 0.13$. This result is not intuitive. At a given spatial resolution, increasing the fluidization velocity leads to a less well-resolved flow field. One possible explanation could be related to the collision model. Indeed, the lower the fluidization velocity, the

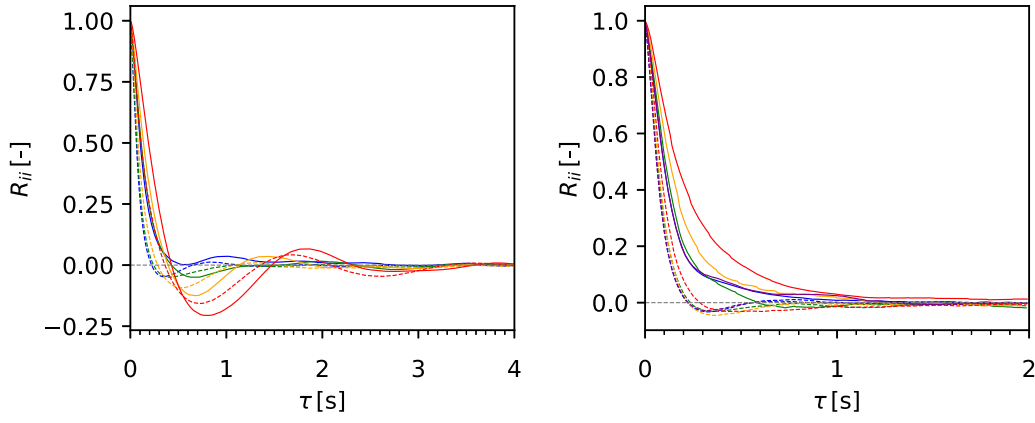


Fig. 19. Particle velocity autocorrelation coefficient. $\frac{D_x}{\Delta x} = 12$ (left), $\frac{D_x}{\Delta x} = 24$ (right). On the right, the grid resolution of $\frac{D_x}{\Delta x} = 36$ is represented in purple for $U_F = 0,073 \text{ m s}^{-1}$ (—). — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. — : vertical component, - - : average of the two horizontal components.

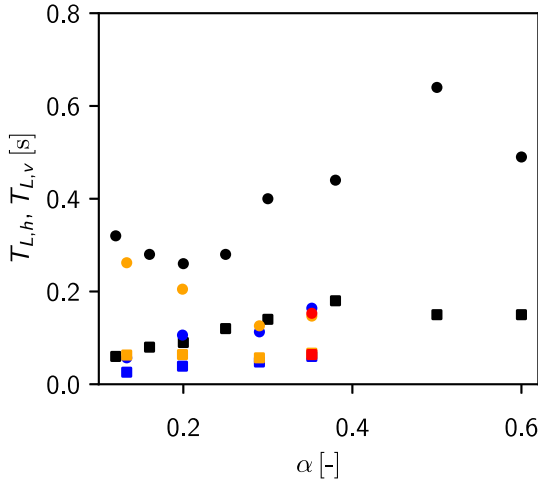


Fig. 20. Lagrangian time scale computed by integrating the autocorrelation coefficient. ■ (resp. •) represents the horizontal (resp. vertical) direction. Color scale indicates experimental or numerical case : • : experimental work of Corona (2008), • : $\frac{D_x}{\Delta x} = 12$, • : $\frac{D_x}{\Delta x} = 24$, • : $\frac{D_x}{\Delta x} = 36$. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

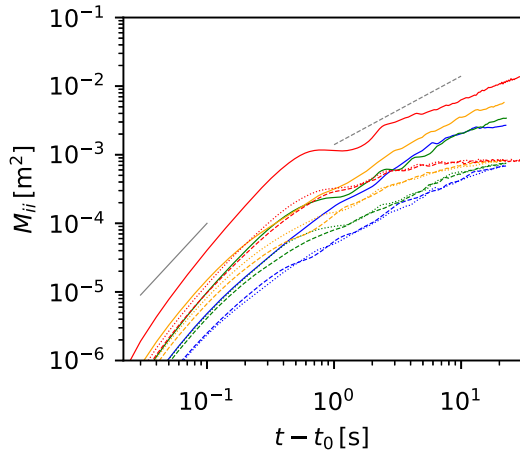


Fig. 21. Variance of the particle trajectory ($\frac{D_x}{\Delta x} = 12$). — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. — : vertical component, - - : horizontal component. — : second-order slope, — : first-order slope.

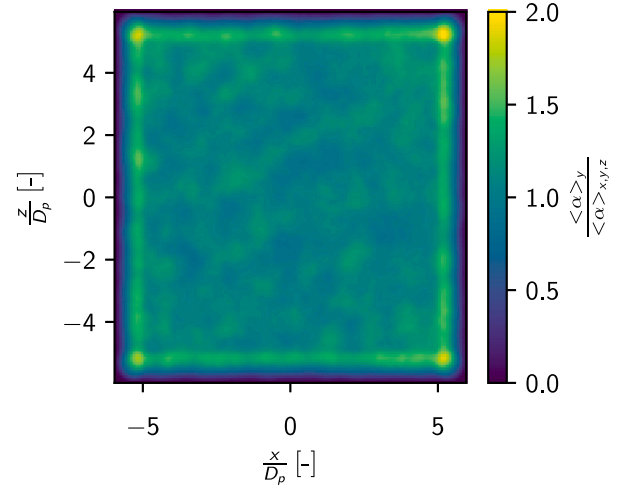


Fig. 22. Temporal averaged map of the solid fraction in the cross-sectional area. $\frac{D_x}{\Delta x} = 12$, $U_F = 0.15 \text{ m s}^{-1}$.

higher the solid fraction, and thus the higher the collision frequency. If the collision model spreads the collision duration excessively, or if collisions are poorly resolved, then the collision frequency within the bed would be affected. Consequently, the transition time between the ballistic and diffusive regimes would be modified, thereby altering particle self-diffusion in the bed.

4.5.3. Characteristic time scales

The Lagrangian diffusion time scale of particles is confronted to other characteristic times of the flow, as the relaxation time:

$$\tau_{fp} = \frac{\rho_p}{\rho_p - \rho_f} \frac{U_F}{(1 - \alpha)g} \quad (21)$$

and the characteristic time of fluid motion at the particle scale:

$$\tau_f = \frac{D_p(1 - \alpha)}{U_F} \quad (22)$$

The computed times are reported in Table 4. The Lagrangian diffusion time scale of the particles in the vertical direction $t_{L,v}$ is always greater than the other characteristic time scales of the flow by a factor of at least 5. It can be seen that the relaxation time of the particles τ_{fp} increases with the fluidization velocity, while the characteristic time of the fluid τ_f decreases. The characteristic diffusion time of the particles in the horizontal direction $t_{L,h}$ is of the same order of magnitude as τ_{fp} and τ_f and is almost constant.

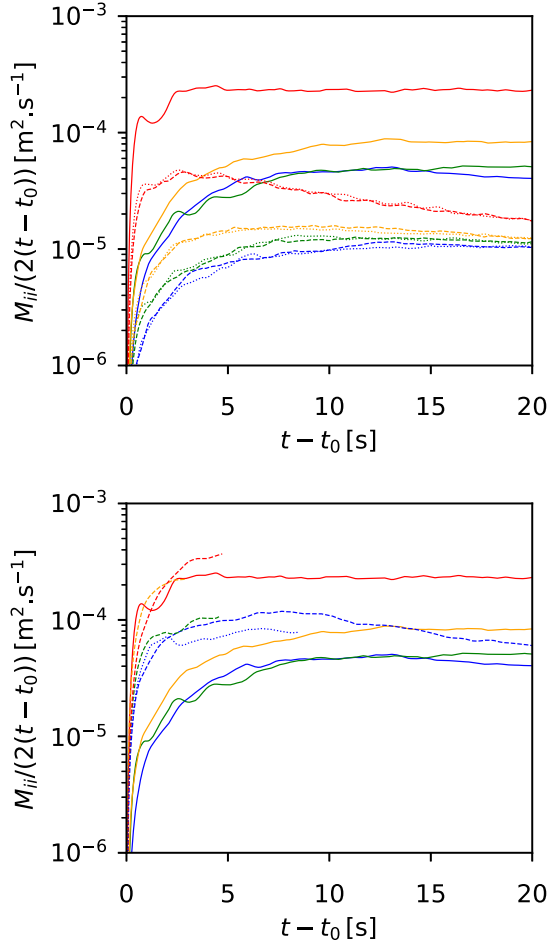


Fig. 23. Self-diffusion coefficient computation. — : $U_F = 0.073 \text{ m s}^{-1}$, — : $U_F = 0.090 \text{ m s}^{-1}$, — : $U_F = 0.120 \text{ m s}^{-1}$, — : $U_F = 0.150 \text{ m s}^{-1}$. Top: — : vertical component, — : horizontal component (along x), ... : horizontal component (along z). Bottom: effect of the grid resolution on the vertical component. — : $\frac{D_x}{\Delta x} = 12$, — : $\frac{D_x}{\Delta x} = 24$, ... : $\frac{D_x}{\Delta x} = 36$.

Table 3

Self-diffusion coefficients. FAC: coefficient computed from the autocorrelation function. VD: coefficient computed from the variance of displacement. $D_h = (D_{xx} + D_{zz})/2$: self-diffusion coefficient in the horizontal direction, $D_v = D_{yy}$: self-diffusion coefficient in the vertical direction. $D_{h,c}$, $D_{v,c}$: experimental data of Corona (2008).

α [–]		0.35		0.29		0.20		0.13	
$\frac{D_x}{\Delta x}$		12	24	12	24	12	24	12	24
D_h [$10^{-5} \text{ m}^2 \text{ s}^{-1}$]	FAC	2.39	2.74	2.94	3.57	3.95	5.81	4.34	6.95
	VD	1.00	–	1.08	–	1.23	–	1.15	–
D_v [$10^{-5} \text{ m}^2 \text{ s}^{-1}$]	FAC	11.5	11.0	12.3	14.9	20.4	37.4	26.1	64.4
	VD	3.89	–	4.92	–	8.44	–	2.00	–
α [–]		0.38		0.25		0.20		0.12	
$D_{h,c}$ [$10^{-5} \text{ m}^2 \text{ s}^{-1}$]	FAC	4.42		6.65		6.86		5.19	
	VD	4.53		7.39		7.19		5.40	
$D_{v,c}$ [$10^{-5} \text{ m}^2 \text{ s}^{-1}$]	FAC	29.2		40.0		51.7		68.0	
	VD	36.1		48.8		51.4		84.2	

5. Eulerian statistics of the fluid and particles agitation

The previous section characterized the agitation of particles within the fluidized bed. In this section, the agitation of the fluid and the particles is characterized using Eulerian data.

Table 4

Characteristic timescales of the flow (in seconds).

$\langle \bar{\alpha} \rangle$	U_F [m s^{-1}]	τ_{fp}	τ_f	$t_{L,h}$	$t_{L,v}$
0.35	0.073	0.031	0.051	0.063	0.262
0.29	0.090	0.035	0.047	0.064	0.205
0.20	0.120	0.041	0.040	0.057	0.126
0.13	0.150	0.047	0.035	0.067	0.147

5.1. Velocity profile

The evolution of the velocity profile along the height of the bed is represented in Fig. 24(a). At the inlet, the average fluid velocity is equal to the fluidization velocity (i.e., the fluid injection velocity). Within the bed, the velocity is almost constant, with a very slightly parabolic profile for the lowest fluidization velocities. The average fluid velocity in the bed is equal to $\frac{U_F}{1-\alpha}$. Above the bed, the velocity decreases to the fluidization velocity. At the inlet and outlet of the bed, slight oscillations are observed for the two lowest fluidization velocities. Mesh refinement indicates that the fluid velocity is overestimated for coarse meshes. Thus, even though very slight differences are observed for the solid fraction between the different grid resolutions (see Figs. 9 and 25). Fig. 24(a) revealed that these differences are not negligible in the fluid velocity profile at the center of the bed. It can be seen that for the case $U_F = 0.15 \text{ m s}^{-1}$, the velocity profile is not sufficiently converged to give an accurate representation of the average velocity at the center of the bed. Finally, the oscillations of the average profile of the particle distribution in the cross-section area (see Figs. 12(b) and 24(b)) are also present in the fluid velocity profiles. At the bed outlet, the average velocity profile is not strictly horizontal, because the average velocity is computed only on vertical slices with a thickness equivalent to two particle diameters at the center of the bed.

5.2. Volumic phase indicator function profile

The solid fraction was previously computed using the Lagrangian method with knowledge of the particles' positions. Here, it is determined using the Eulerian method with the phase indicator function (see Fig. 25). Qualitatively, the curves are identical to those determined using the Lagrangian method. In the figure on the right, the influence of the grid resolution is more clearly identified, indicating better convergence in time for the Eulerian profile. Although the oscillatory behavior is identical for both resolutions, the solid fraction at the center of the bed is slightly higher for the fine resolution than for the coarse one. This trend is observed at all four fluidization velocities.

5.3. Difference between Eulerian and Lagrangian particle velocity variances

The behavior of particles can be characterized according to Eulerian and Lagrangian approaches. The Lagrangian average captures only the translational velocity, while the Eulerian mean captures both the rotational velocity and the translational velocity. The Eulerian average should therefore result in higher fluctuating kinetic energy. Differences between both approaches then appear and must be quantified. For this purpose, the standard deviation of the horizontal and vertical velocities of the particles is investigated. Fig. 26 reports the ratio between the standard deviation of the particle velocity computed using the Eulerian method and the standard deviation computed using the Lagrangian method. On average, the deviations associated with the horizontal component are greater than those associated with the vertical component. Furthermore, the deviations are almost always greater than 1, indicating that the standard deviation of the velocity computed using the Eulerian method is greater than that computed using the Lagrangian method, as expected. For the grid resolution of 12 meshes per particle diameter, the deviations range from 16% to 18% for the horizontal component and from 5.0% to 11% for the vertical component. For 24 meshes per particle diameter, these deviations fall below 11% for the horizontal component and below 5.4% for the vertical component.

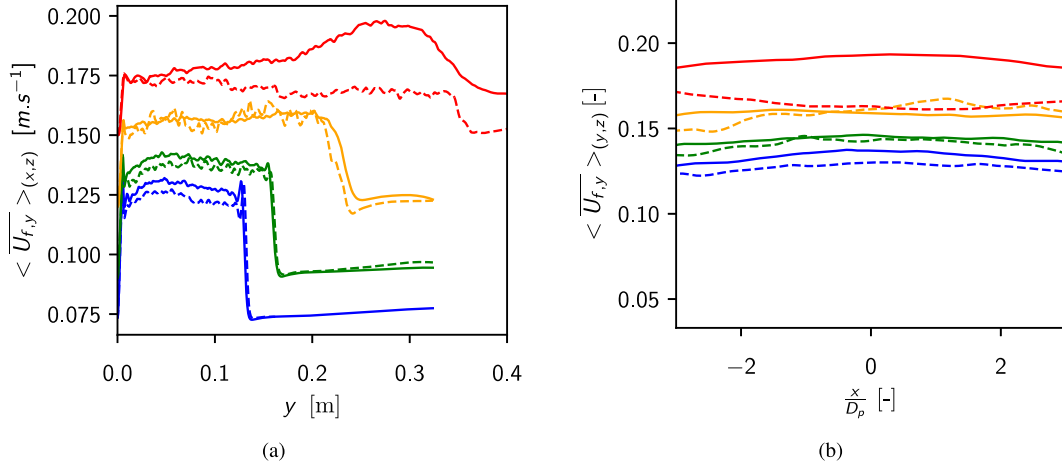


Fig. 24. (a) Time averaged velocity profile as a function of the bed height. (b) Time averaged velocity profile as a function of the wall distance. — : $U_F = 0,073 \text{ m.s}^{-1}$, — : $U_F = 0,090 \text{ m.s}^{-1}$, — : $U_F = 0,120 \text{ m.s}^{-1}$, — : $U_F = 0,150 \text{ m.s}^{-1}$. — : $\frac{D_p}{\Delta x} = 12$. - - : $\frac{D_p}{\Delta x} = 24$.

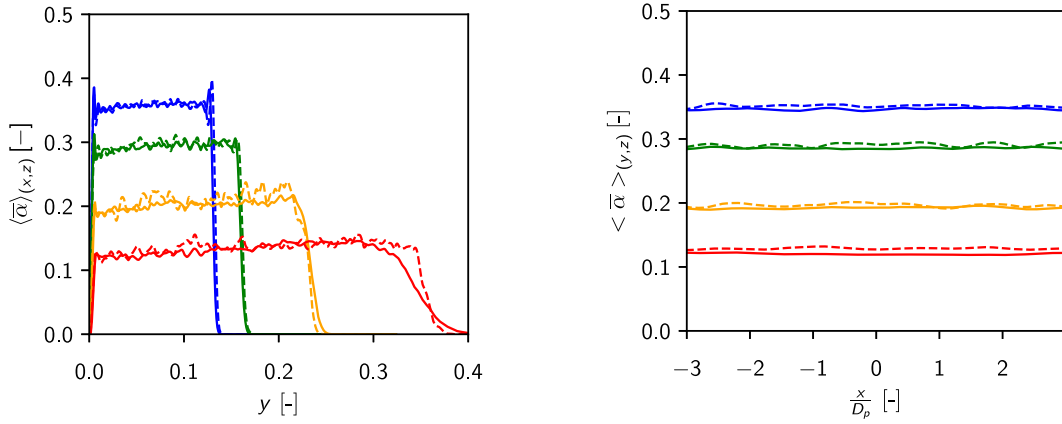


Fig. 25. Eulerian computation of the solid fraction. — : $U_F = 0,073 \text{ m.s}^{-1}$, — : $U_F = 0,090 \text{ m.s}^{-1}$, — : $U_F = 0,120 \text{ m.s}^{-1}$, — : $U_F = 0,150 \text{ m.s}^{-1}$. — : $\frac{D_p}{\Delta x} = 12$. - - : $\frac{D_p}{\Delta x} = 24$.

5.4. Fluctuating kinetic energy

The fluctuating kinetic energies of the fluid and particles are computed as follows:

$$E_p = \frac{1}{2} \sum_i \overline{\langle u'_{p,i,L} \rangle_p^2} \quad (23)$$

$$E_f = \frac{1}{2} \sum_i \overline{\langle u'_{f,i} \rangle_{(x,y,z)}^2} \quad (24)$$

The kinetic energy of the fluid as a function of height in the bed is represented in Fig. 27. For cases with $U_F \leq 0.12 \text{ m.s}^{-1}$, the fluid turbulent kinetic energy increases along the bed height. However, its amplitude is relatively small as it never exceeds 12%. It then decreases exponentially at the bed outlet to reach zero, indicating that the SPT is negligible compared to the BIF and that the fluctuating kinetic energy in the bed is solely due to the BIF. For larger fluidization velocities, the E_f increase is more pronounced. In the case where $U_F = 0.15 \text{ m.s}^{-1}$, the value of the fluid's kinetic energy is multiplied by 1.8 between the inlet and outlet of the bed, then decreases at the outlet. For the finest grid resolution, a kind of plateau in the turbulent kinetic energy in the bed is observed. Furthermore, the computed kinetic energy is always greater for this resolution except for the highest fluidization velocity, for which 12 meshes per diameter does not allow to capture the small scales features of the flow.

In Fig. 28, the turbulent kinetic energy of the particle velocity is compared to that of the fluid. For both phases, the kinetic energy decreases with the solid fraction, which is consistent with experimental data. The slope is well predicted for the kinetic energy of the particles, but underestimated for the kinetic energy of the fluid. One possible explanation is the difference in the geometry of the cross-section area. The grid refinement only has an effect for the highest fluidization velocity and helps to reduce the discrepancy with the experimental results. Finally, the ratio between the kinetic energy of the fluid and that of the particles increases with solid fraction. An order of magnitude difference is observed between the two quantities for the highest solid fraction.

Increasing the grid resolution is expected to improve the representation of turbulent kinetic energy (particularly at the highest fluidization velocity), as well as the computation of particle velocity autocorrelation coefficients and the self-diffusion coefficient, thereby providing more reliable estimates of the Lagrangian time scale.

6. Extension of Alm  ras et al. (2019) model

6.1. Proposed model

The analysis of the variance of particle and fluid velocity enables the evaluation of the model developed by Alm  ras et al. (2019). It relates the variance of the fluctuating fluid velocity to the average presence

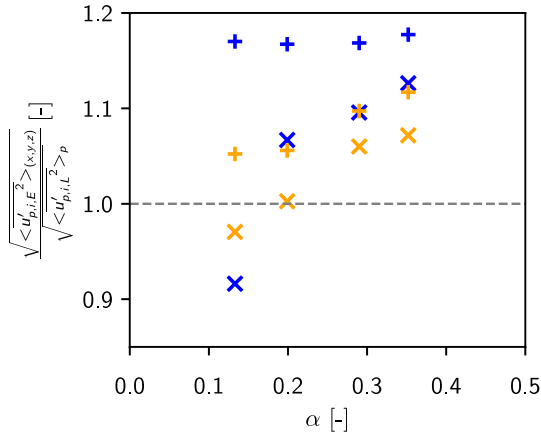


Fig. 26. Ratio between the standard deviation of particle velocity computed with an Eulerian approach ($\sqrt{\langle u'_{p,i,E}{}^2 \rangle_{(x,y,z)}}$) and the standard deviation computed with the Lagrangian approach ($\sqrt{\langle u'_{p,i,L}{}^2 \rangle_p}$). + : horizontal component of the velocity, x : vertical component of the velocity. — : $\frac{D_p}{\Delta x} = 12$, — : $\frac{D_p}{\Delta x} = 24$.

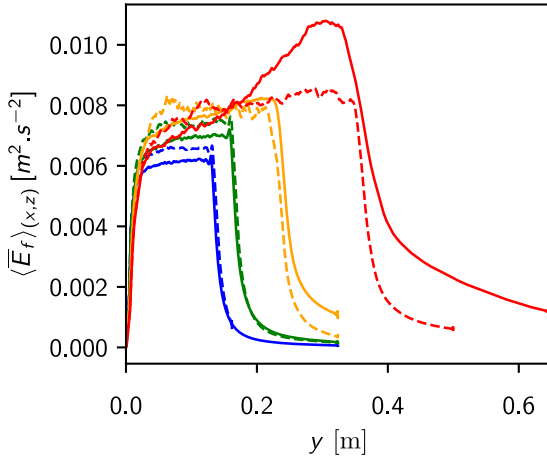


Fig. 27. Fluctuating kinetic energy of the fluid as a function of the bed height. — : $U_F = 0,073 \text{ m s}^{-1}$, — : $U_F = 0,090 \text{ m s}^{-1}$, — : $U_F = 0,120 \text{ m s}^{-1}$, — : $U_F = 0,150 \text{ m s}^{-1}$. — : $\frac{D_p}{\Delta x} = 12$, — : $\frac{D_p}{\Delta x} = 24$.

rate in the flow and to the average fluid velocity according to Eq. (3). Fig. 29(a) confronts the experimental results of Corona (2008) and Alm  ras et al. (2019) with the numerical results of Ozel et al. (2017) and present results. First, one notes that the total fluctuating kinetic energy (symbol +) coincides with the numerical results of Ozel. On the other hand, a discrepancy is observed with the experimental results of Corona. Contrary to the numerical results, the experimental points do not describe a linear model for which the slope is the coefficient γ . Furthermore, the slopes obtained from the experimental results of Alm  ras et al. (2019) differ from those obtained by numerical simulation of the Corona configuration (Corona, 2008). In fact, in the present case, $\gamma_v^2 = 2.31$ and $\gamma_h^2 = 0.95$ are found, compared to $\gamma_v^2 = 1.54$ and $\gamma_h^2 = 0.63$ in the study by Alm  ras et al. (2019). To extend the validity of Eq. (3), it is proposed to add a dependency on particle diameter and density ratio through the Stokes number:

$$\overline{\langle u'_{i,i}{}^2 \rangle_{(x,y,z)}} = \zeta_i \frac{\langle \bar{U}_{i,y} \rangle_{(x,y,z)}^2}{St_i} \alpha (1 - \alpha) \quad (25)$$

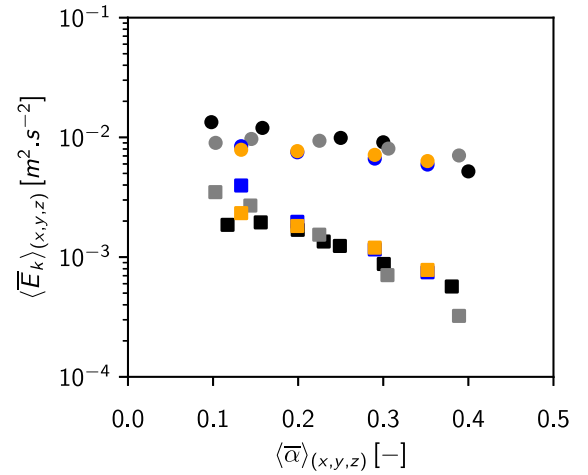


Fig. 28. Fluctuating kinetic energy of the fluid as a function of the solid fraction. ■ : particles (computed with the lagrangian approach), • : fluid. — : experimental work of Corona (2008), — : numerical work of Ozel et al. (2017), — : $\frac{D_p}{\Delta x} = 12$, — : $\frac{D_p}{\Delta x} = 24$.

where ζ_i is a parameter to be determined and $St_i = \frac{8\rho_p}{3\rho_f C_D}$ is the Stokes number associated with the terminal sedimentation velocity of an isolated particle in the fluid under consideration. Alm  ras et al. (2021) determined experimentally that the latter equals 0.22 m s^{-1} . The drag coefficient is computed by a weight/buoyancy balance on an isolated particle:

$$C_D^f = \frac{4}{3} \frac{\rho_p - \rho_f}{\rho_f U_i^2} g D_p \quad (26)$$

where U_i is the settling velocity in the fluid. Noting C and A as the indices referring to the experimental studies by Corona (2008) and Alm  ras et al. (2019), one obtains $St_{i,C} = 5.26$ and $St_{i,A} = 7.35$. One can relate ζ_v and ζ_h to the coefficient proposed by Alm  ras et al. (2019): $\gamma_{v,A}^2 \simeq \frac{\zeta_v}{St_{i,A}}$ and $\gamma_{h,A}^2 \simeq \frac{\zeta_h}{St_{i,A}}$. In the original version of the correlation proposed by Alm  ras et al. (2019), the velocity variance is only a function of the void fraction and the mean fluid velocity. However, for a given void fraction, the influence of particles on fluid agitation varies with their radius and density. The Stokes number therefore accounts for both the particle response time and its sensitivity to the surrounding fluid flow.

6.2. Assessment of the scaling law

Fig. 29(b) shows the correlation (see Eq. (25)) where the coefficients ζ_i were computed using the least squares method. This yields $\zeta_v = 11.7$ and $\zeta_h = 4.82$. The proposed correlation allows to predict the variance of fluid velocity fluctuations within a 20% confidence interval. Further simulations in different configurations are required to assess the validity of the new correlation. The choice of the new configurations is based on several criteria. First, the settling velocity need to be known precisely, based on an experimental measurement. Indeed, an error δu on the settling velocity would involve an error $(\delta u)^2$ on the velocity variance. Then, the Reynolds and Stokes number should be on the same order of magnitude than the previous configurations. Few experimental studies on liquid–solid fluidized beds satisfying these criteria are available in the literature. Therefore, the experimental results of Mordant and Pinton (2000) on the measurement of the settling velocity of a single particle in an infinite medium are used. The fluid is water at ambient temperature. Two configurations were selected: a particle of diameter 0.5 mm and density 2560 kg m^{-3} , and a particle of diameter 1.5 mm with the same density. The physical

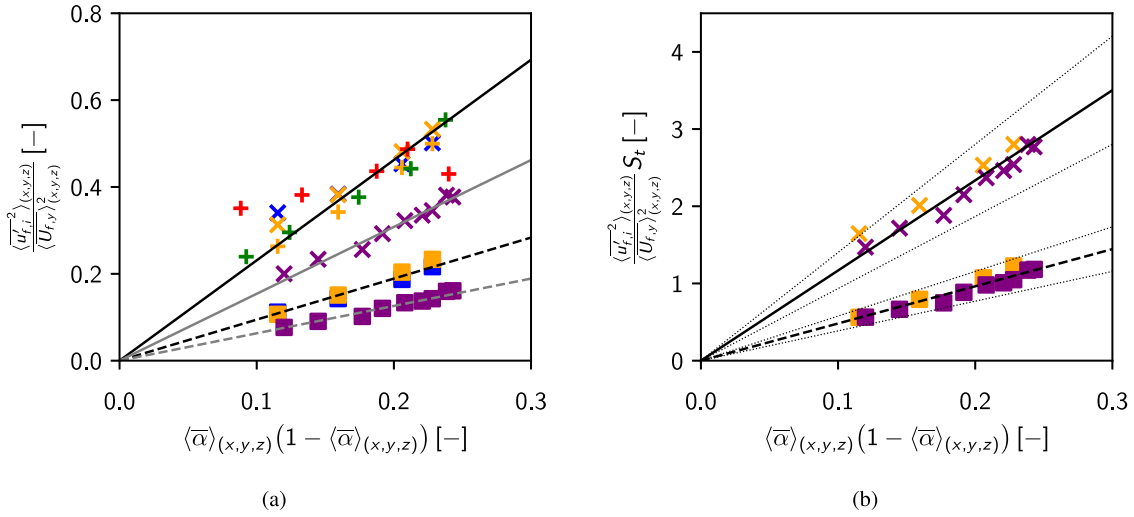


Fig. 29. Fluctuating kinetic energy variance as a function of the solid fraction. (a) Model of Alm  ras et al. (2019). (b) new correlation. + : total energy, $\times$: vertical component, $\blacksquare$: horizontal component. Colors: $\color{red}$: experimental work of Corona (2008), $\color{green}$: numerical work of Ozel et al. (2017), $\color{purple}$: experimental work of Alm  ras et al. (2019), $\color{blue}$: $\frac{D_p}{\Delta x} = 12$, $\color{orange}$: $\frac{D_p}{\Delta x} = 24$.

Line styles: (a). --- , --- : correlation (3) of Alm  ras et al. (2019) with modified γ_i for the vertical and horizontal component of the velocity. --- , --- : original correlation (3) of Alm  ras et al. (2019) for the vertical and horizontal component of the velocity. (b): --- , --- : correlation (25) for the horizontal and the vertical component of the velocity. $\cdots$: 15% confident interval. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Table 5
Physical parameters of the studied configurations.

Case	D_p [mm]	U_t [m s ⁻¹]	$\frac{\rho_p}{\rho_f}$	Re_t	St_t
Corona (2008)	6.0	0.240	1.59	530	5.26
Alm��ras et al. (2019)	4.0	0.220	1.53	523	7.35
Case 1 (Mordant and Pinton, 2000)	0.5	0.074	2.56	42.2	3.67
Case 2 (Mordant and Pinton, 2000)	1.5	0.218	2.56	367	10.6

parameters of the experiment of Corona (2008) and Alm  ras et al. (2019) and the two new configurations are synthesized in Table 5. For each new configuration, several simulations were performed with different fluidization velocities. For the case 1 of Table 5, the selected fluidization velocities are 0.018 m s⁻¹, 0.025 m s⁻¹, and 0.030 m s⁻¹. For the case 2 of Table 5, the selected fluidization velocities are 0.049 m s⁻¹ and 0.090 m s⁻¹. To save computation resources, the simulations were performed with a grid resolution of $\frac{D_p}{\Delta x} = 12$. The time evolution of the bed height is represented in Fig. 30(a). For each configuration, few fluctuations are observed in the time evolution of the bed height. The solid fraction is between 0.2 and 0.4, which is very close to previous studied configurations.

The new correlation is evaluated in Fig. 31. For case 1, the variance of the fluctuating kinetic energy is well predicted by the new scaling law (shown in blue in the figure). However, the correlation is not suitable for case 2. The calculated variance of the kinetic energy is greatly underestimated by the correlation. One possible explanation could be the limitation of the correlation's validity for a certain range of terminal Stokes numbers. Indeed, the terminal Reynolds number, density ratio, and particle diameter for case 2 are of the same order of magnitude as for the other configurations studied. The only difference lies in the Stokes number. One possible explanation could be that for Stokes numbers greater than 10, collisions between particles play an important role in the energy transfer. The observed difference could correspond to the missing portion of energy transfers in the model. In the present study, all particle Stokes numbers remain below 10.6. According to Legendre et al. (2006), the collision Stokes number can therefore be associated with a nearly zero wet coefficient of restitution. The wet restitution coefficient characterizes the fluid-induced damping of particle kinetic energy during collisions. A value of zero indicates

that colliding particles do not rebound, whereas a value of one corresponds to elastic-like collisions with no energy loss due to viscous effects. In the present conditions, since the Stokes number is below 10, the wet restitution coefficient is close to zero. As a result, collisions do not generate rebound-induced chaotic motion, and particle dynamics are mainly governed by advection by the fluid. Under this regime, an increase in the Stokes number indicates that particles become more resistant to the surrounding fluid motion and may disrupt fluid-induced agitation. Consequently, for Stokes numbers below 10, an increase in the Stokes number is expected to lead to a decrease in the fluid velocity variance. For the case 2 in Fig. 31, the corresponding Stokes number is 10.6. At this value, the wet restitution coefficient begins to increase, and the proposed extension is no longer valid. One may suggest that beyond this threshold, the fluid velocity variance begins to increase with the Stokes number due to the increasingly chaotic motion of particles. A further extension of the correlation would be required to account for collision effects on the fluid velocity variance at Stokes numbers greater than 10. An in-depth investigation of the collision frequency would be required to assess the hypothesis.

7. Conclusion

Particle-resolved simulations of liquid–solid fluidized beds were performed with the one-fluid Navier–Stokes formulation of TrioCFD, which uses a Front-Tracking method to locate the surface of particles. The experimental study of Corona (2008) was examined numerically. The macroscopic behavior of the flow (bed height, solid fraction) was characterized. The influence of grid resolution and viscosity model was quantified. With the chosen parameters, the Richardson–Zaki law was recovered. Then, an in-depth analysis of the Lagrangian statistics of the bed was performed. The spatial distribution of the particles, analyzed using the radial pair distribution, showed the homogeneous nature of the fluidized bed. Analysis of the particle velocity established that the particles tend to rise in the center of the bed and fall back near the wall. The probability density functions of particle velocity fluctuations indicated a symmetric and perfectly Gaussian distribution of the horizontal component. A rightward shift was observed for the vertical component, in agreement with Corona (2008). The study of the autocorrelation coefficient of particle velocity highlighted the strong

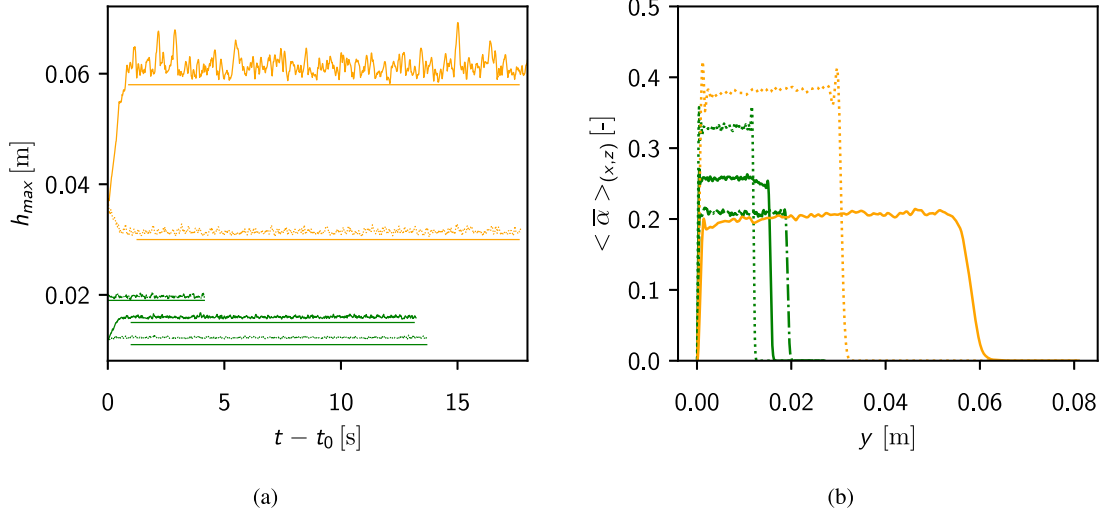


Fig. 30. (a) Temporal evolution of the bed height. (b) Temporal average solid fraction as a function of the bed height. —: Case 1, $U_F = 0.025 \text{ m s}^{-1}$ (solid lines), $U_F = 0.018 \text{ m s}^{-1}$ (dotted lines), $U_F = 0.030 \text{ m s}^{-1}$ (dashed-dotted lines). —: Case 2, $U_F = 0.090 \text{ m s}^{-1}$ (solid lines), $U_F = 0.049 \text{ m s}^{-1}$ (dotted lines). Horizontal lines represent the period for computing averages.

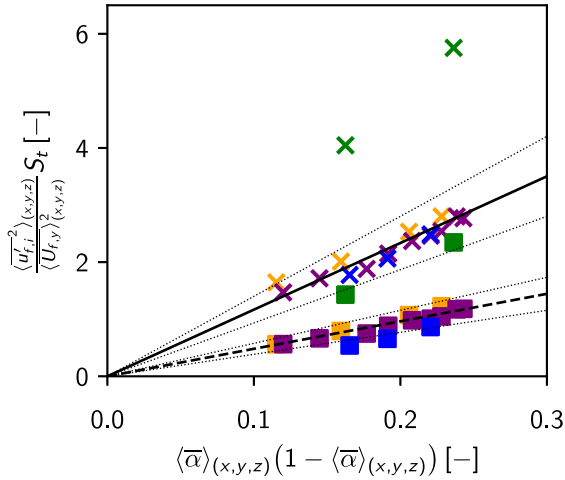


Fig. 31. New correlation of the fluctuating kinetic energy variance as a function of the solid fraction. $\times$: vertical component, $\blacksquare$: horizontal component. Colors: — : experimental work of Alm  ras et al. (2019), — : present numerical results of the configuration of Corona (2008) with $\frac{p_z}{\Delta x} = 24$. — : Case 1 of Table 5, — : Case 2 of Table 5, Line styles: —, — : correlation (25) for the horizontal and the vertical component of the velocity. . . . : 15% confident interval. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

influence of grid resolution on the recovery of the Lagrangian time scale. The influence of grid resolution was also remarkable on the time-averaged velocity profile as a function of bed height for the highest fluidization velocity and on the fluctuating kinetic energy profile. Finally, Eulerian statistics on fluid velocity were used to evaluate the Almer  s correlation (Alm  ras et al., 2019). The linear behavior of the correlation is well reproduced for the configuration studied, but with a different coefficient. In order to generalize the law, a dependence on the Stokes number based on the terminal velocity was proposed. The new correlation was evaluated on a new set of particle-resolved simulations. The correlation was validated for small particle-to-fluid density ratios and for Reynolds numbers based on terminal velocity up to 530 and Stokes numbers based on the terminal velocity up to 7. The correlation may be employed within the Euler–Euler framework to close the drag force model. Further study would be required to extend the correlation

to higher Stokes numbers. Moreover, an investigation into the influence of geometry on the particle and fluid statistics would help to clarify the slight discrepancies observed with the experimental data. Finally, to enable industrial applications, the influence of heat and mass transfer on the particle and fluid statistics must be quantified in order to extend the correlation to such cases.

CRediT authorship contribution statement

E. Butaye: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **S. Mer:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization. **F. Bataille:** Writing – review & editing. **A. Toutant:** Writing – review & editing, Visualization, Validation, Supervision, Methodology, Investigation, Formal analysis, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Time convergence and grid sensitivity of the Lagrangian velocity profiles of particles

The time convergence of the Lagrangian velocity profiles of the particles is evaluated. Let Δt_{avg} be the simulation duration over which the averages are computed. Let A_1 , A_2 and A_3 the averages computed respectively over the first half of Δt_{avg} , the second half of Δt_{avg} and the entire duration Δt_{avg} . The profiles are shown in Fig. A.32. Time convergence is achieved for the two lowest fluidization speeds. However, differences between the various averages on the profile curve at

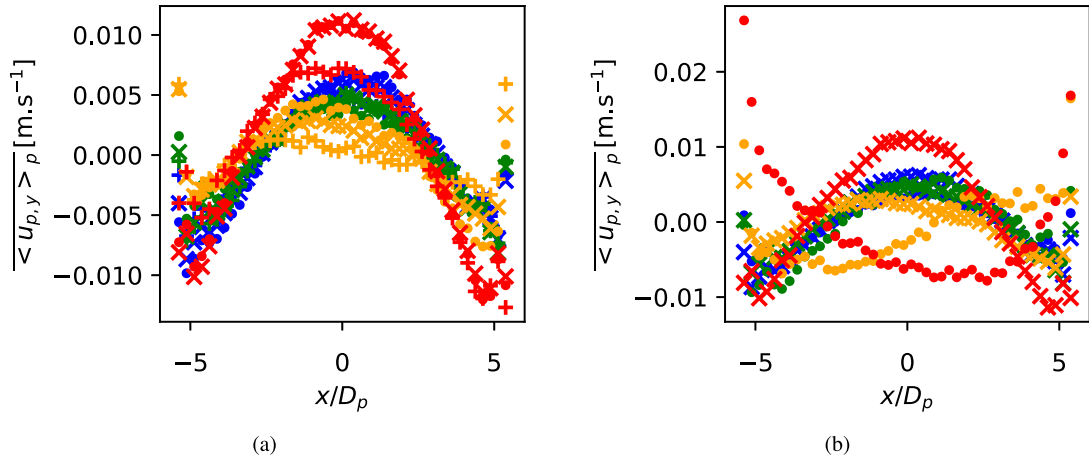


Fig. A.32. Vertical velocity profile of solid particles computed in a Lagrangian manner. $\times$: $U_F = 0,073 \text{ m s}^{-1}$, $\times$: $U_F = 0,090 \text{ m s}^{-1}$, $\times$: $U_F = 0,120 \text{ m s}^{-1}$, $\times$: $U_F = 0,150 \text{ m s}^{-1}$. (a) Time convergence: $\cdot$: A_1 , $+$: A_2 , $\times$: A_3 . (b) Grid sensitivity: $\times$: $\frac{D_p}{dx} = 12$, $\cdot$: $\frac{D_p}{dx} = 24$.

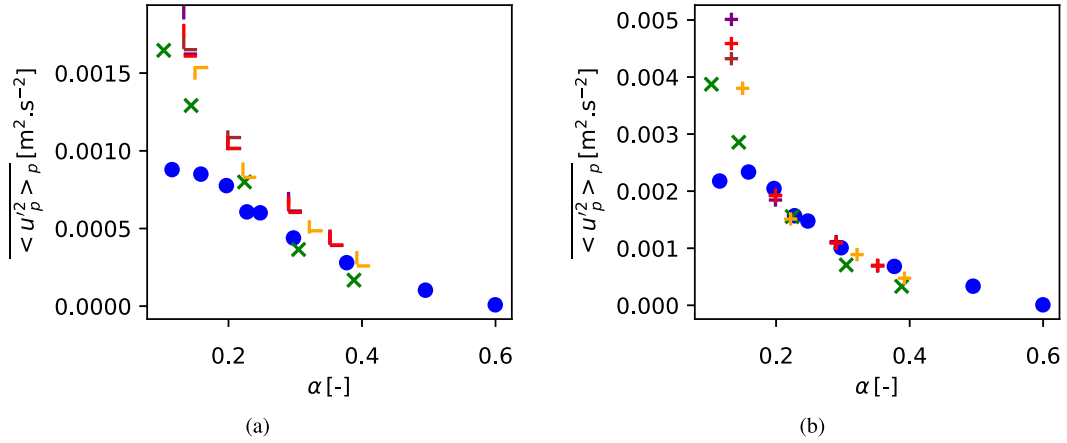


Fig. B.33. Variance of horizontal (left) and vertical (right) components of particle velocity for $\frac{D_p}{dx} = 12$. $\cdot$: experimental data of Corona (2008). $\times$: numerical data of Ozel et al. (2017), $+$: numerical data (Hamidi et al., 2023), $—$: average over the first half of the simulation time, $—$: average over the second half of the simulation time, $—$: average over the entire simulation time.

the center of the domain are observed for the two highest fluidization speeds.

Fig. A.32 compares the profiles obtained for a resolution of 12 meshes per diameter with those obtained for a resolution of 24 meshes per diameter. For the two highest fluidization velocities, the simulation time is not long enough to provide a good representation of the average profile at 24 meshes per particle diameter. For the lowest fluidization velocity, it can be seen that at the center of the bed, the particle velocity appears to be overestimated at the coarsest grid resolution. This is consistent with the overestimation of friction between the fluid and the particles with coarse grids. For a fluidization velocity of 0.09 m s^{-1} , the particle velocity front is identical for both resolutions, but differences arise near the walls. However, the simulated physical time with the resolution of 24 meshes per diameter is too short to draw any conclusions.

In conclusion, time convergence is ensured for the two lowest fluidization velocities. Longer simulated physical times would be required to ensure convergence of the velocity profiles associated with the highest fluidization velocities. Only the trend is captured: the particles rise at the center of the domain and descend between the wall and a distance of approximately four particle diameters. Regarding grid sensitivity, only the two lowest fluidization velocities can be discussed. The variations in the profile appear to be correctly predicted with the coarse grid. Further simulations would be required to quantify the discrepancies observed for the two highest fluidization velocities. The

slow convergence of the Lagrangian velocity profiles can be explained by the fact that a large number of particles must be captured in each slice, and that there are only 2134 of these particles in the whole computation domain, located at specific points by their gravity center.

Appendix B. Time convergence of the velocity variance

The convergence of the variance of the horizontal and vertical components of velocity is assessed. Fig. B.33 shows the averages computed over the first half of the simulation time ($—$), over the second half ($—$) and over the entire simulation time ($—$). It can be seen that the variances have converged for velocities less than or equal to 0.12 m s^{-1} . However, for $U_F = 0.15 \text{ m s}^{-1}$, slight differences are observed between the three averages, indicating that the simulation has not yet converged in time. One should point out that the average values are computed over 25 s for the three lowest fluidization velocities and over 40 s for the highest fluidization velocity, which is longer than in the studies by Hamidi et al. (2023) (5 s) or Ozel et al. (2017) (10 s).

Appendix C. Convergence of the ensemble average of velocity autocorrelations

The convergence of the ensemble average of velocity autocorrelations is investigated. It is achieved by varying the number of particles for which the autocorrelation is computed. Figs. C.34 and C.35 show

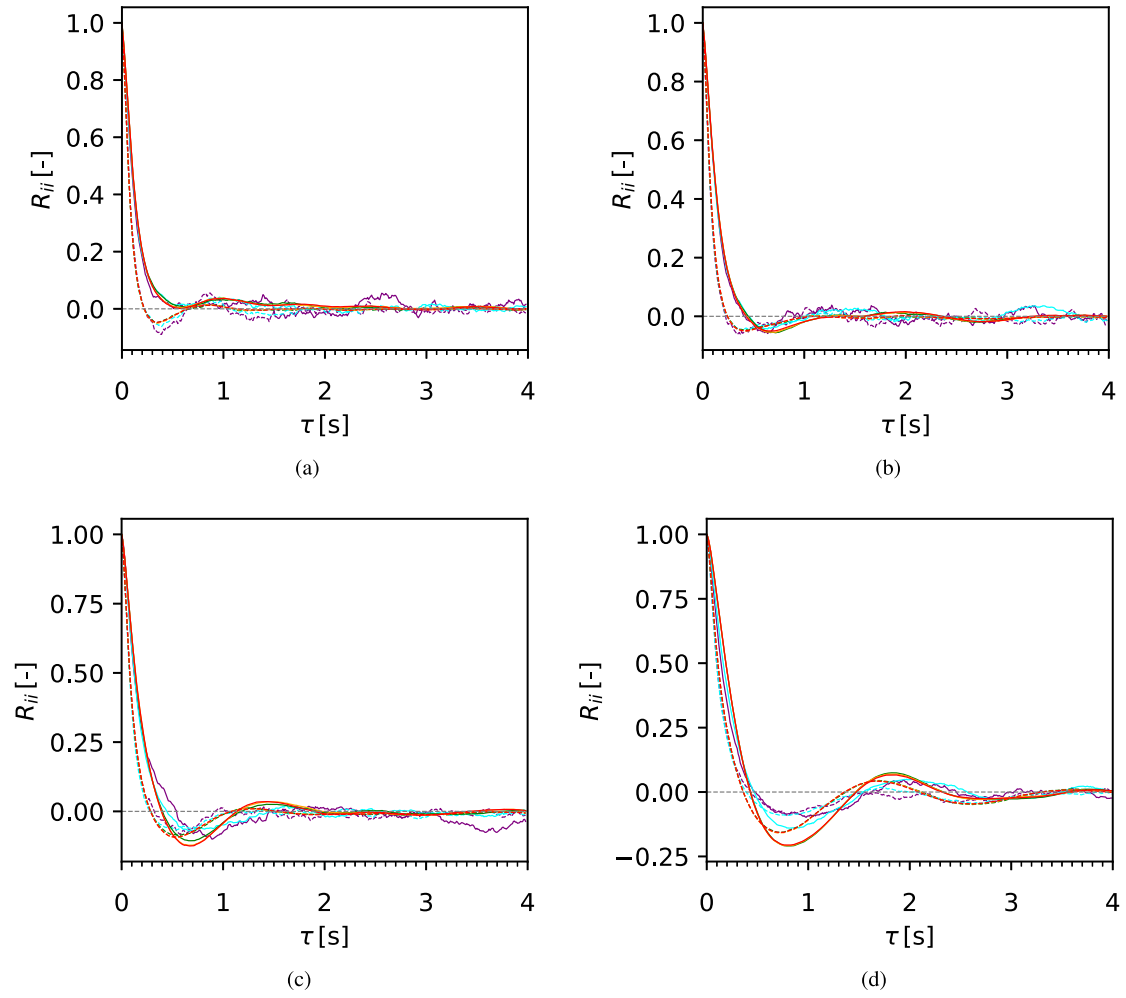


Fig. C.34. Autocorrélation de la vitesse lagrangienne des particules. $\frac{D_p}{\Delta x} = 12$. (a) $U_F = 0.073 \text{ m s}^{-1}$, (b) $U_F = 0.090 \text{ m s}^{-1}$, (c) $U_F = 0.120 \text{ m s}^{-1}$, (d) $U_F = 0.150 \text{ m s}^{-1}$.
 — : $N_p = 12$, — : $N_p = 50$, — : $N_p = 500$, — : $N_p = 1000$, — : $N_p = 2134$.

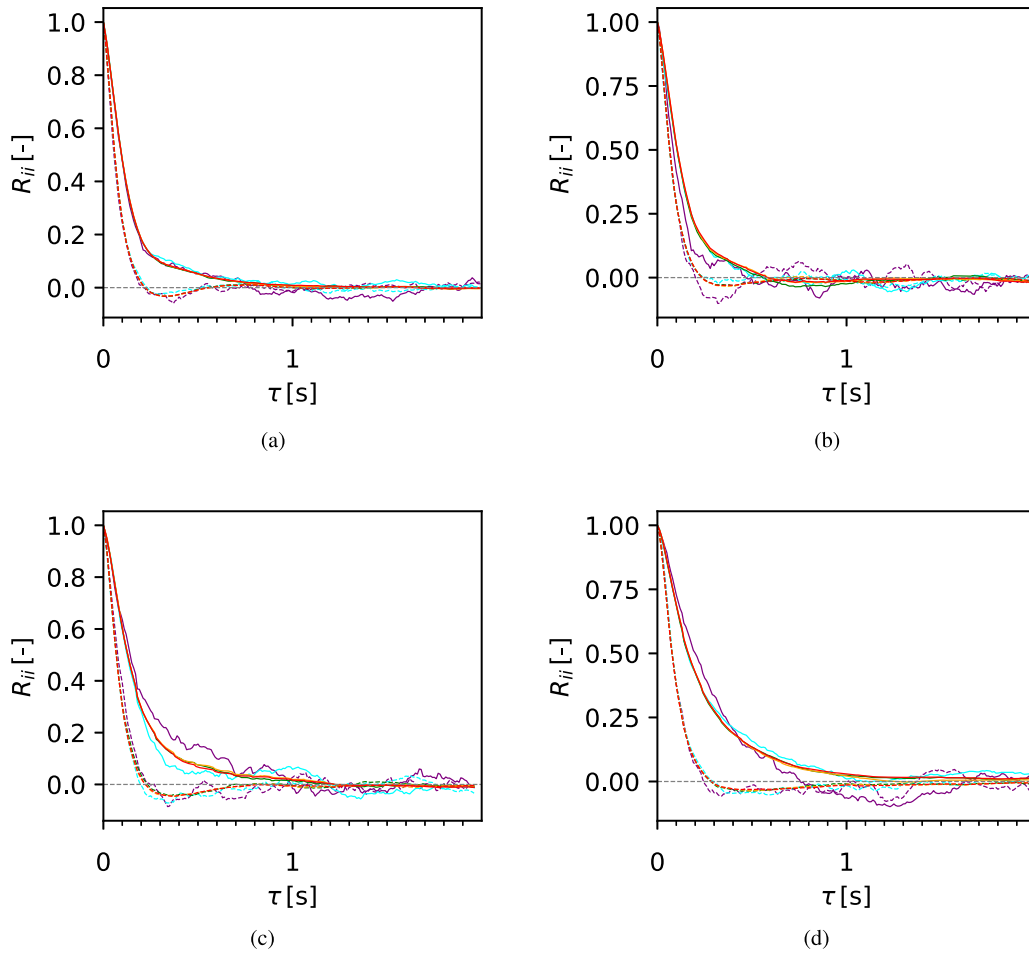


Fig. C.35. Autocorrelation of the Lagrangian particle velocity. $\frac{D_p}{\Delta x} = 24$. (a) $U_F = 0.073 \text{ m s}^{-1}$, (b) $U_F = 0.090 \text{ m s}^{-1}$, (c) $U_F = 0.120 \text{ m s}^{-1}$, (d) $U_F = 0.150 \text{ m s}^{-1}$.
 — : $N_p = 12$, — : $N_p = 50$, — : $N_p = 500$, — : $N_p = 1000$, — : $N_p = 2134$.

the convergence for a grid resolution of 12 and 24 meshes per particle diameter, respectively. In both cases, convergence is achieved for $N_p = 1000$. However, slight differences are observed between the curves for $N_p = 500$ and $N_p = 1000$. Thus, for future studies, 500 particles can be used to compute these coefficients.

Data availability

Data will be made available on request.

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